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ECE 340
THE ANALYTICAL BITS

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A note on how this book was made. The Analytical Bits originated from my ECE 340 lectures. I used ChatGPT to extract annotations from my lecture notes, add brief explanations alongside the equations, and typeset the content in LaTeX. I then revised the resulting LaTeX files, added figures created with Circuit Macros, and compiled them into ECE 340: The Analytical Bits. Think of this as a companion resource—not a replacement for lectures, the full lecture notes, or the reference book

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To my college professors–Paul Panayotatos,

S. J. Orfanidis, Sigrid McAfee, and M. F. Caggiano–

for their rigor, clarity, and high standards.

1 Diode

1.1 Small-Signal Analysis

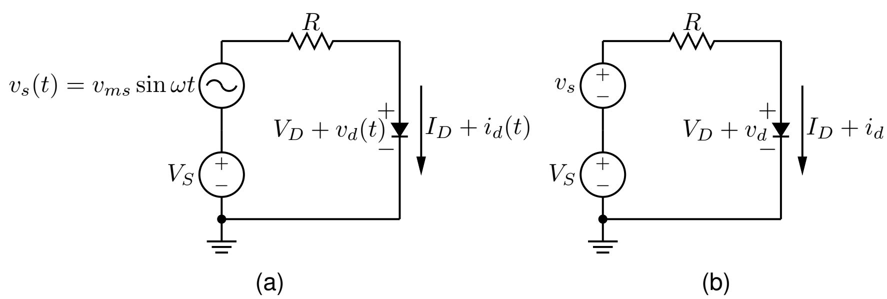

Figure 1.1: (a) Small-signal analysis. (b) Small-signal analysis using simplified notation.

Let $v_D(t)$ and $i_D(t)$ denote the total diode voltage and current, respectively. Then

$$v_D(t) = V_D + v_d(t), \qquad i_D(t) = I_D + i_d(t).$$

Here, v_s denotes the small-signal source voltage associated with the instantaneous source voltage $v_s(t)$, and i_d denotes the small-signal diode current associated with the instantaneous diode current $i_D(t)$. Figure 1.1(a) can therefore be redrawn as Fig. 1.1(b).

$$v_D = V_D + v_d, \qquad i_D = I_D + i_d.$$

Use the diode equation, including the ideality factor n:

$$i_D = I_S\left(e^{\frac{v_D}{nV_T}} - 1\right) \approx I_S e^{\frac{v_D}{nV_T}} \quad \text{(forward bias)}.$$

Substituting $v_D = V_D + v_d$ gives

$$i_D \approx I_S e^{\frac{V_D+v_d}{nV_T}} = I_S e^{\frac{V_D}{nV_T}} e^{\frac{v_d}{nV_T}}. \tag{1.1}$$

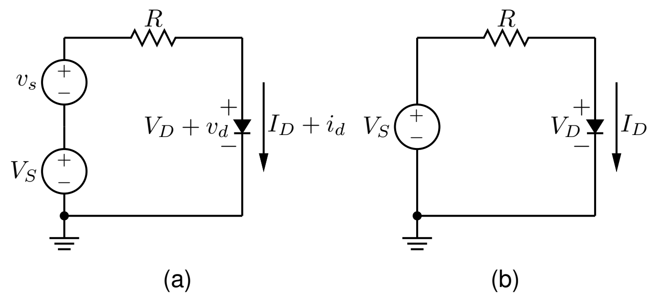

Figure 1.2: (a) Complete circuit. (b) DC circuit obtained by setting v_s, v_d, and i_d to zero.

For small v_d (that is, $v_d \ll nV_T$), use the first-order Taylor approximation

$$e^x \approx 1 + x,$$

with $x = \dfrac{v_d}{nV_T}$:

$$\begin{aligned} i_D &\approx I_S e^{\frac{V_D}{nV_T}} \left(1 + \frac{v_d}{nV_T}\right) \\ &= \underbrace{I_S e^{\frac{V_D}{nV_T}}}_{I_D} + \underbrace{\left(\frac{I_D}{nV_T}\right) v_d}_{i_d}. \end{aligned} \tag{1.2}$$

Hence, the key small-signal relation is

$$i_d = g_m v_d, \qquad g_m \triangleq \frac{I_D}{nV_T}. \tag{1.3}$$

1.1.1 *Separation of DC and AC Quantities*

1.1.1.1 *DC Analysis*

Apply KVL to the circuit in Fig. 1.1(b):

$$V_S + v_s - (I_D + i_d)R = V_D + v_d. \tag{1.4}$$

The DC equivalent circuit of Fig. 1.1(b) is obtained by setting v_s to zero and, consequently, setting i_d to zero. Therefore,

$$V_D = V_S - I_D R. \tag{1.5}$$

1.1.1.2 *Small-Signal Analysis*

The small-signal component of Eq. 1.4 is obtained by setting the DC quantities V_S, V_D, and I_D to zero. This gives

$$v_s - i_d R = v_d. \tag{1.6}$$

Using (1.2)–(1.3), the AC equation can also be written as

$$v_d = v_s - g_m v_d R \quad \Longrightarrow \quad v_d = \frac{v_s}{1 + g_m R} = \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R} v_s \tag{1.7}$$

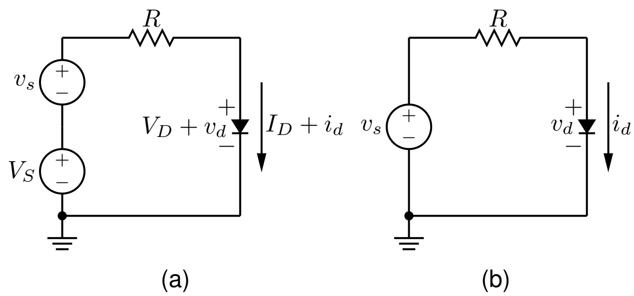

Figure 1.3: (a) Complete circuit. (b) Small-signal (AC) circuit obtained by setting V_S, V_D, and I_D to zero.

1.1.1.3 Small-Signal Equivalent Circuit Model

Equation 1.7 shows that v_d can be found using the voltage-divider relation. Therefore, in the small-signal model, the diode can be replaced by an equivalent resistance of $1/g_m$, as shown in Fig. 1.4.

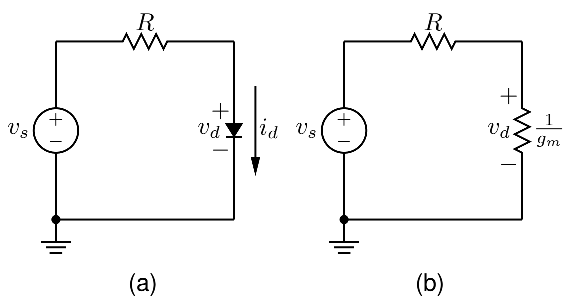

Figure 1.4: (a) Small-signal circuit. (b) Small-signal equivalent circuit with the diode replaced by a resistor.

2
BJT

2.1 Small-Signal Analysis

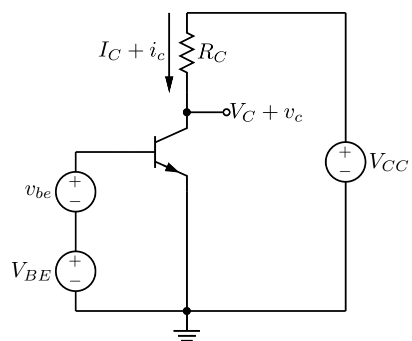

Figure 2.1: Small-signal analysis

We consider a BJT biased at the DC operating point (V_{BE}, V_C, I_C) and perturbed by a small-signal voltage v_{be}. v_{be} and v_c are small-signal voltages representing the instantaneous voltages of $v_{be}(t)$ and $v_c(t)$. i_c is a small-signal current representing the instantaneous current ($i_c(t)$). The total base-emitter voltage and collector current can be written as

$$v_{BE} = V_{BE} + v_{be}, \qquad i_C = I_C + i_c.$$

Ignoring the Early effect, the collector current is

$$i_C = I_S e^{\frac{V_{BE}+v_{be}}{V_T}} = I_S e^{\frac{V_{BE}}{V_T}} e^{\frac{v_{be}}{V_T}} \tag{2.1}$$

$$= I_C \, e^{\frac{v_{be}}{V_T}}. \tag{2.2}$$

If $v_{be} \ll V_T$, we may use the first-order approximation

$$e^x \approx 1 + x \qquad (x \ll 1),$$

with $x = \frac{v_{be}}{V_T}$. Substituting this approximation into the previous expression gives

$$i_C \approx I_C \left(1 + \frac{v_{be}}{V_T}\right) = I_C + \underbrace{\frac{I_C}{V_T} v_{be}}_{i_c}. \tag{2.3}$$

Therefore, the small-signal collector current is

$$i_c \approx \frac{I_C}{V_T} v_{be}. \tag{2.4}$$

We define the transconductance as

$$g_m \triangleq \frac{I_C}{V_T}. \tag{2.5}$$

Thus, the small-signal collector current is proportional to the small-signal base-emitter voltage:

$$i_c = g_m v_{be}.$$

2.2 *Small-Signal Analysis with Early Effect*

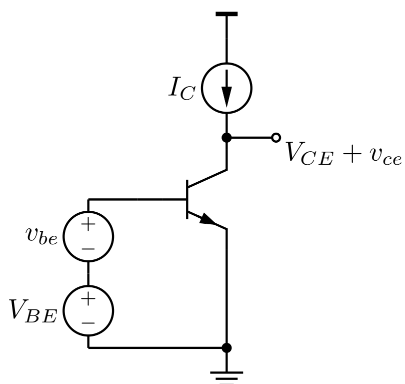

Figure 2.2: Small-signal analysis with Early effect included.

With Early effect, the collector current may be modeled as

$$i_C = I_S e^{\frac{V_{BE}+v_{be}}{V_T}} \left(1 + \frac{V_{CE} + v_{ce}}{V_A}\right), \tag{2.6}$$

where V_A is the Early voltage and

$$v_{CE} = V_{CE} + v_{ce}.$$

Linearize about the operating point assuming small signals:

$$v_{be} \ll V_T, \qquad v_{ce} \ll V_A.$$

First, separate DC and small-signal factors:

$$i_C = I_S e^{\frac{V_{BE}}{V_T}} e^{\frac{v_{be}}{V_T}} \left(1 + \frac{V_{CE}}{V_A} + \frac{v_{ce}}{V_A}\right). \tag{2.7}$$

Using $e^{\frac{v_{be}}{V_T}} \approx 1 + \frac{v_{be}}{V_T}$ and dropping second-order products $\left(\frac{v_{be}}{V_T}\right)\left(\frac{v_{ce}}{V_A}\right)$, we obtain

$$i_C \approx I_S e^{\frac{V_{BE}}{V_T}} \left(1 + \frac{V_{CE}}{V_A}\right) + I_S e^{\frac{V_{BE}}{V_T}} \left(1 + \frac{V_{CE}}{V_A}\right) \frac{v_{be}}{V_T} + I_S e^{\frac{V_{BE}}{V_T}} \frac{v_{ce}}{V_A}. \tag{2.8}$$

Recognize the DC operating current

$$I_C = I_S e^{\frac{V_{BE}}{V_T}} \left(1 + \frac{V_{CE}}{V_A}\right), \tag{2.9}$$

so the incremental current is

$$i_c \approx \frac{I_C}{V_T} v_{be} + \frac{I_C}{V_A} v_{ce} = g_m v_{be} + \frac{v_{ce}}{r_o}, \tag{2.10}$$

where

$$g_m = \frac{I_C}{V_T}, \qquad r_o \triangleq \frac{V_A}{I_C}. \tag{2.11}$$

$i_c = 0$ since the collector current is constant:

$$0 \approx i_c = g_m v_{be} + \frac{v_{ce}}{r_o} \quad \Longrightarrow \quad v_{ce} \approx -g_m r_o\, v_{be}. \tag{2.12}$$

2.3 Small-Signal Parameters

2.3.1 Definition of Transconductance g_m

Assume the (DC) collector current is given by the exponential model (ignoring Early effect for the moment):

$$I_C = I_S e^{V_{BE}/V_T}. \tag{2.13}$$

We define the small-signal transconductance as

$$g_m \triangleq \frac{i_c}{v_{be}}. \tag{2.14}$$

Because small-signal quantities are infinitesimal perturbations around the bias point, the ratio is equivalently the derivative:

$$\frac{i_c}{v_{be}} = \lim_{\Delta V_{BE} \to 0} \frac{\Delta I_C}{\Delta V_{BE}} = \frac{dI_C}{dV_{BE}}. \tag{2.15}$$

Differentiate $I_C = I_S e^{V_{BE}/V_T}$ with respect to V_{BE}:

$$\frac{dI_C}{dV_{BE}} = \frac{d}{dV_{BE}} \left(I_S e^{V_{BE}/V_T}\right) = I_S e^{V_{BE}/V_T} \left(\frac{1}{V_T}\right) = \frac{I_C}{V_T}. \tag{2.16}$$

Therefore,

$$\boxed{g_m = \frac{I_C}{V_T}}. \tag{2.17}$$

Assumption: For this derivation we assume I_C is (approximately) not a function of V_{CE} (i.e., we neglect the Early effect). The Early effect is incorporated later via r_o.

2.3.2 *Determination of r_π*

Bias the BJT at (I_B, I_C) and apply small-signal perturbations (i_b, i_c) due to a small v_{be}.

Using the definition of current gain,

$$\beta \triangleq \frac{I_C}{I_B}, \tag{2.18}$$

and assuming the same proportionality holds for small signals around the operating point,

$$\beta = \frac{i_c}{i_b}. \tag{2.19}$$

From $i_c = g_m v_{be}$ and $i_b = i_c/\beta$, we get

$$i_b = \frac{i_c}{\beta} = \frac{g_m}{\beta} v_{be}. \tag{2.20}$$

Thus the small-signal resistance seen looking into the base-emitter port is

$$r_\pi \triangleq \frac{v_{be}}{i_b} = \frac{v_{be}}{(g_m/\beta)v_{be}} = \frac{\beta}{g_m}. \tag{2.21}$$

So,

$$\boxed{r_\pi = \frac{\beta}{g_m}}. \tag{2.22}$$

2.3.3 *Determination of Output Resistance r_o*

Include the Early effect:

$$I_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right), \tag{2.23}$$

where V_A is the Early voltage.

Define the small-signal output resistance as

$$r_o \triangleq \frac{v_{ce}}{i_c}. \tag{2.24}$$

Holding V_{BE} fixed (so the exponential factor is constant at the operating point), the small-signal collector current due to a change in V_{CE} is

$$i_c = \frac{\partial I_C}{\partial V_{CE}} v_{ce}. \tag{2.25}$$

Compute the partial derivative:

$$\frac{\partial I_C}{\partial V_{CE}} = I_S e^{V_{BE}/V_T}\left(\frac{1}{V_A}\right) = \frac{I_C}{V_A}, \tag{2.26}$$

where in the last step we use that at the operating point $I_C \approx I_S e^{V_{BE}/V_T}$ (or more precisely, $I_C/(1 + V_{CE}/V_A)$ depending on convention; the usual small-signal result is the same to first order).

Therefore,

$$r_o = \frac{v_{ce}}{i_c} = \frac{1}{\partial I_C/\partial V_{CE}} = \frac{V_A}{I_C}. \tag{2.27}$$

So,

$$\boxed{r_o = \frac{V_A}{I_C}}. \tag{2.28}$$

2.4 Summary

All three key small-signal parameters depend on the bias collector current I_C:

$$\boxed{g_m = \frac{I_C}{V_T}}, \qquad \boxed{r_\pi = \frac{\beta}{g_m}}, \qquad \boxed{r_o = \frac{V_A}{I_C}}. \tag{2.29}$$

2.5 DC and Small-Signal Equivalent Circuits

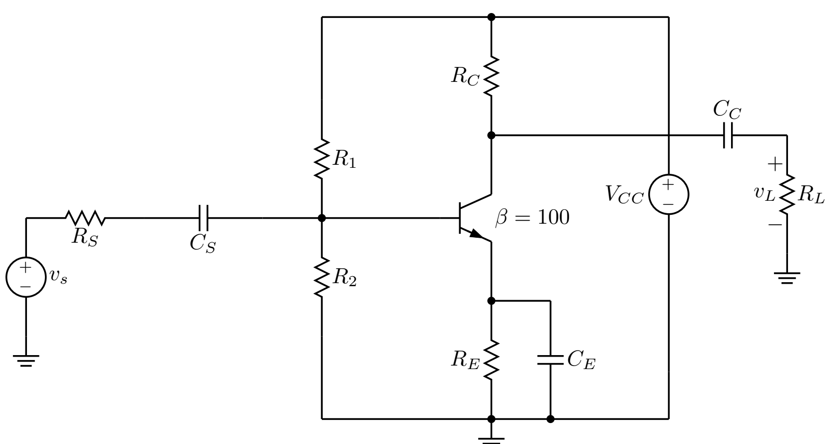

Figure 2.3: A common-emitter amplifier circuit with emitter degeneration.

Use the following guidelines to draw the DC equivalent circuit:

1. *Set all independent AC sources to zero.*
   For a voltage source, this means *short* the source: $v_s \to 0$ (replace it with a wire).
   (If there were an independent current source, you would *open* it.)

2. *Replace all capacitors with opens (capacitors block DC).*
   Open C_S, C_C, and C_E.

3. *Clean up the schematic by removing any components that become disconnected by the opened capacitors.*
   For example, the load R_L is disconnected through C_C and therefore does not affect the DC bias.

4. *Keep the DC supply and the biasing resistor/transistor network.*

5. *Sanity check: identify DC current paths.*
   Every node voltage you intend to solve for at DC should have a clear resistive path to the supply and/or ground.

   Figure 2.4 shows the DC equivalent circuit of Fig. 2.3.

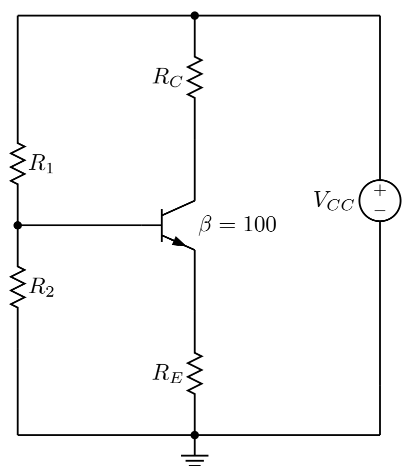

Figure 2.4: The DC equivalent circuit.

Use the following guidelines to draw the small-signal circuit:

1. *Set independent DC sources to zero (AC ground).*
   For a DC voltage source, this means *short* it: $V_{CC} \to 0$ (the supply node becomes AC ground). For a DC current source, this means *open* it.

2. *Replace coupling and bypass capacitors with shorts.*
   Short C_S, C_C, and C_E.

3. *Combine resistors that now share nodes due to the shorts and AC grounds:*

   - With V_{CC} shorted, the top of R_C is at AC ground.
   - With C_C shorted, R_L connects directly to the collector node.
   - Therefore, the collector load is often written as $R_C \parallel R_L$ (and in parallel with r_o if included).
   - With C_E shorted, the emitter node is (approximately) at AC ground, so R_E is bypassed for AC.
   - With C_S shorted, the source v_s drives the base node through R_S.

- The bias divider becomes a base-to-ground resistance, $R_1 \parallel R_2$.

4. **Sanity check:** identify what is grounded and what is floating. At midband, coupling capacitors should not leave the signal path open; instead, they should connect the source and load into the AC network.

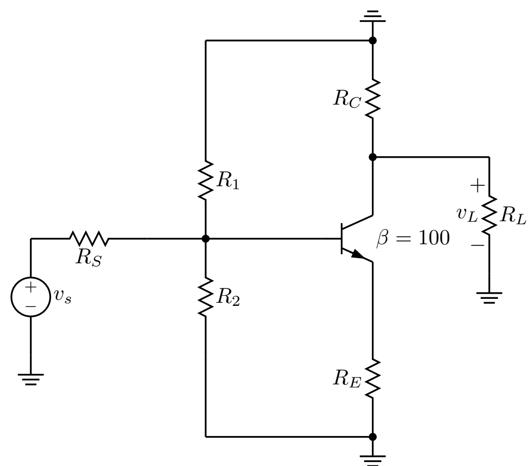

Figure 2.5: The small-signal circuit.

The transistor in Fig. 2.5 can be replaced by its small-signal model:

- r_π is between the base and emitter.
- The dependent current source $g_m v_\pi$ produces current from collector to emitter.
- r_o is between the collector and emitter. Include r_o when $V_A \neq \infty$, and omit it when $V_A = \infty$.

Figure 2.6 shows the small-signal equivalent circuit of Fig. 2.5.

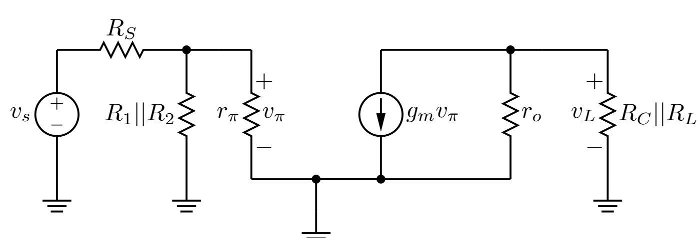

Figure 2.6: The small-signal equivalent circuit.

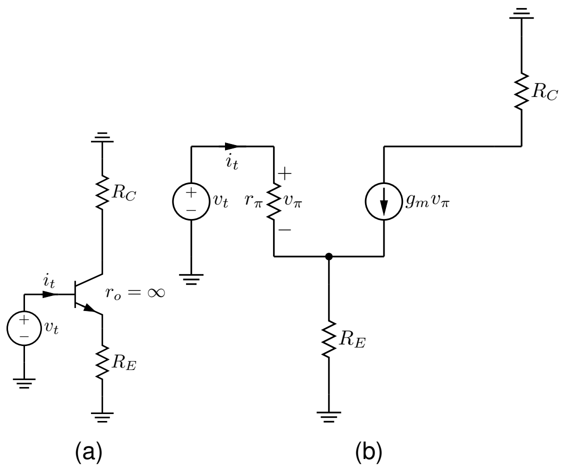

Figure 2.7: Small-signal resistance looking into the base. (a) Test circuit used to determine the base resistance. (b) Small-signal circuit with the transistor replaced by its small-signal model.

2.6 *Small-Signal Resistance*

2.6.1 *Base resistance*

A test source is applied at the input to compute the base resistance, as shown in Fig. 2.7(a).

$$R_\mathrm{b} \triangleq \frac{v_t}{i_t}.$$

Applying KCL at the emitter node in Fig. 2.7(b) yields

$$i_t + g_m v_\pi = \frac{v_t - v_\pi}{R_E}.$$

Multiply by R_E and rearrange:

$$i_t R_E + g_m v_\pi R_E + v_\pi = v_t.$$

Since the test current enters the base and flows through r_π, we identify $i_t = i_\pi$; hence $v_\pi = i_t r_\pi$. Substituting gives

$$\begin{aligned} i_t R_E + g_m (i_t r_\pi) R_E + i_t r_\pi &= v_t, \\ i_t (R_E + g_m r_\pi R_E + r_\pi) &= v_t. \end{aligned}$$

Therefore,

$$\begin{aligned} R_\mathrm{b} = \frac{v_t}{i_t} &= R_E + g_m r_\pi R_E + r_\pi \\ &= r_\pi (1 + g_m R_E) + R_E. \end{aligned}$$

Using $g_m r_\pi \approx \beta$, the result is written as

$$\boxed{R_\mathrm{b} \approx r_\pi + (\beta + 1) R_E.} \tag{2.30}$$

2.6.2 *Emitter Resistance*

Apply a test voltage v_t at the emitter node and define the test current i_t as the current entering the circuit at that node as shown in Fig. 2.8(a). The base is connected to ground through R_B (i.e., the bias network seen from the base is replaced by R_B). The transistor is replaced by its small-signal hybrid-π model as shown in Fig. 2.8(b) with r_π from base to emitter and a controlled current source $g_m v_\pi$ from collector to emitter. We assume $r_o = \infty$.

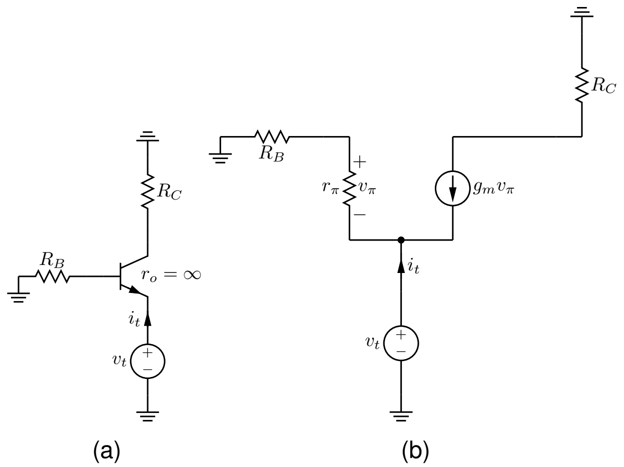

Figure 2.8: Small-signal resistance looking into the emitter. (a) Test circuit used to determine the emitter resistance. (b) Small-signal circuit with the transistor replaced by its small-signal model.

Writing KCL at the emitter terminal (sum of currents leaving the emitter node equals zero) gives

$$i_t + g_m v_\pi + \frac{0 - v_t}{R_B + r_\pi} = 0. \tag{2.31}$$

The current through the series combination $(R_B + r_\pi)$ from the emitter node to ground is

$$i_{(R_B + r_\pi)} = \frac{0 - v_t}{R_B + r_\pi}. \tag{2.32}$$

That same current flows through r_π, so

$$v_\pi = i_{(R_B+r_\pi)}\, r_\pi = \left(\frac{0 - v_t}{R_B + r_\pi}\right) r_\pi = -\frac{v_t\, r_\pi}{R_B + r_\pi}. \tag{2.33}$$

Substitute (2.55) into (2.31):

$$i_t + g_m \left(-\frac{v_t r_\pi}{R_B + r_\pi}\right) + \frac{-v_t}{R_B + r_\pi} = 0, \tag{2.34}$$

$$i_t - \frac{g_m r_\pi\, v_t}{R_B + r_\pi} - \frac{v_t}{R_B + r_\pi} = 0, \tag{2.35}$$

$$i_t = \frac{v_t}{R_B + r_\pi}\left(1 + g_m r_\pi\right). \tag{2.36}$$

Therefore,

$$R_e \triangleq \frac{v_t}{i_t} = \frac{R_B + r_\pi}{1 + g_m r_\pi}. \tag{2.37}$$

Using the hybrid-π identity

$$\beta = g_m r_\pi \qquad \Longleftrightarrow \qquad r_\pi = \frac{\beta}{g_m}, \tag{2.38}$$

rewrite (2.37):

$$R_e = \frac{R_B + r_\pi}{1 + \beta} = \frac{R_B}{\beta + 1} + \frac{r_\pi}{\beta + 1}. \tag{2.39}$$

Since $\dfrac{r_\pi}{\beta + 1} = \dfrac{\beta / g_m}{\beta + 1}$, for $\beta \gg 1$ we have

$$\frac{r_\pi}{\beta + 1} \approx \frac{1}{g_m}. \tag{2.40}$$

Thus the commonly used approximation is

$$\boxed{R_e \approx \frac{R_B}{\beta + 1} + \frac{1}{g_m}, \qquad (\beta \gg 1).} \tag{2.41}$$

2.6.3 *Collector Resistance*

Determine the small-signal resistance looking into the collector by applying a test voltage v_t at the collector node as shown in Fig. 2.9(a) and computing

$$R_c \triangleq \frac{v_t}{i_t}.$$

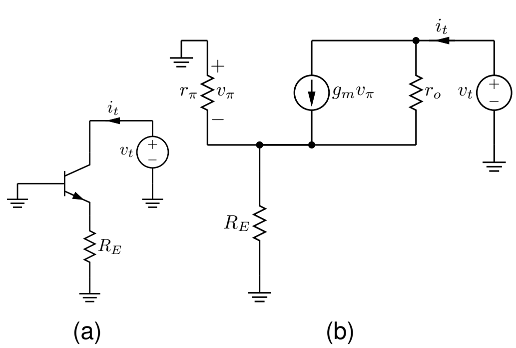

Figure 2.9: Small-signal resistance looking into the collector. (a) Test circuit used to determine the collector resistance. (b) Small-signal circuit with the transistor replaced by its small-signal model.

Replace the BJT by its hybrid-π model including r_π, the dependent current source $g_m v_\pi$, and the output resistance r_o as shown in Fig. 2.9(b). The base is at AC ground, so

$$v_\pi \triangleq v_b - v_e = 0 - v_e = -v_e.$$

The emitter is connected to ground through R_E and through r_π (via the base at AC ground), hence the emitter sees $R_E \parallel r_\pi$ to ground.

At the collector node, the test current i_t splits into the controlled source current and the r_o current.

$$i_t = g_m v_\pi + \frac{v_c - v_e}{r_o} \tag{2.42}$$

$$= g_m v_\pi + \frac{v_t - (-v_\pi)}{r_o} \tag{2.43}$$

$$= g_m v_\pi + \frac{v_t + v_\pi}{r_o}. \tag{2.44}$$

All current that arrives at the emitter from the collector must flow to ground through $R_E \parallel r_\pi$. Since $v_e = -v_\pi$,

$$i_t = \frac{v_e}{R_E \parallel r_\pi} = \frac{-v_\pi}{R_E \parallel r_\pi}, \tag{2.45}$$

so

$$v_\pi = -\, i_t \, (R_E \parallel r_\pi). \tag{2.46}$$

Return to the collector KCL and multiply by r_o:

$$i_t r_o = g_m r_o \, v_\pi + v_t + v_\pi \tag{2.47}$$

$$= v_t + v_\pi(1 + g_m r_o). \tag{2.48}$$

Rearrange:

$$v_t = i_t r_o - v_\pi(1 + g_m r_o). \tag{2.49}$$

Substitute $v_\pi = -i_t(R_E \parallel r_\pi)$:

$$v_t = i_t r_o + i_t(R_E \parallel r_\pi)(1 + g_m r_o). \tag{2.50}$$

Therefore,

$$\boxed{R_c = \frac{v_t}{i_t} = r_o + (1 + g_m r_o)(R_E \parallel r_\pi).} \tag{2.51}$$

2.6.4 *Effective Resistance Between Base and Emitter*

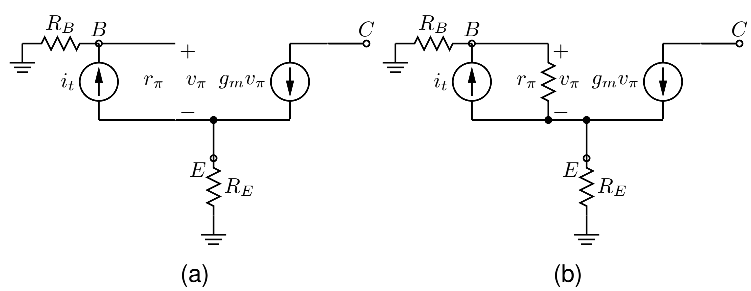

Figure 2.10: Resistance between base and emitter. (a) Without r_π. (b) With r_π.

The goal is to find the effective resistance seen by the test current source i_t between the base and emitter terminals in Fig. 2.10(b).

1. Define the test quantity Let the voltage across the test source be V_t, with polarity positive at the base node and negative at the emitter node. Then

$$V_t = v_b - v_e.$$

Because r_π is connected directly between the same two nodes, it will appear in parallel with the resistance found from the controlled-source path.

2. Express the base voltage The base node is connected to ground through R_B. Since the test current i_t enters the base node, the voltage at the base in Fig. 2.10(a) is

$$v_b = i_t R_B.$$

3. Express the emitter voltage At the emitter node, the current through R_E is the downward current delivered by the controlled source minus the current injected upward by the test source. Thus

$$i_{R_E} = g_m v_\pi - i_t.$$

Therefore,

$$v_e = i_{R_E} R_E = (g_m v_\pi - i_t) R_E.$$

Since v_π is the voltage from base to emitter,

$$v_\pi = v_b - v_e = V_t.$$

So we may write

$$v_e = (g_m V_t - i_t) R_E.$$

4. Form the test resistance expression Substitute the expressions for v_b and v_e into

$$V_t = v_b - v_e :$$

$$V_t = i_t R_B - (g_m V_t - i_t) R_E.$$

Expand the right-hand side:

$$V_t = i_t R_B - g_m V_t R_E + i_t R_E.$$

Move the V_t term to the left side:

$$V_t + g_m R_E V_t = i_t (R_B + R_E).$$

Factor V_t:

$$V_t (1 + g_m R_E) = i_t (R_B + R_E).$$

Hence,

$$\frac{V_t}{i_t} = \frac{R_B + R_E}{1 + g_m R_E}.$$

5. Include the effect of r_π The resistor r_π is connected directly between the same two terminals as the test source, so it is in parallel with the resistance just derived. Therefore, the effective resistance seen by i_t is

$$R_{\text{eff}} = \left(\frac{R_B + R_E}{1 + g_m R_E}\right) \parallel r_\pi.$$

Final result

$$\boxed{R_{\text{eff}} = \left(\frac{R_B + R_E}{1 + g_m R_E}\right) \parallel r_\pi} \tag{2.52}$$

This follows the same logic as the handwritten work: first write $V_t = v_b - v_e$, then express v_b through R_B, express v_e through R_E, use $v_\pi = V_t$, and finally compute V_t/i_t before placing the result in parallel with r_π.

2.6.5 *Examples*

Common-Emitter Amplifier

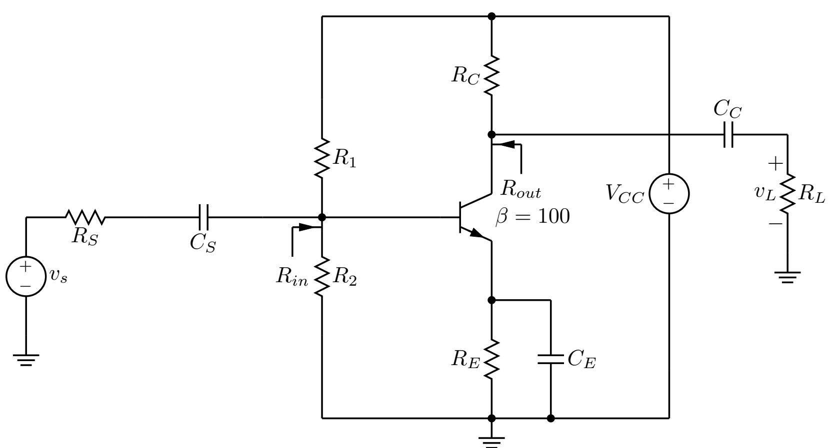

Figure 2.11: Input and output resistance of a common-emitter amplifier.

C_E bypasses R_E at small-signal frequencies. Equation 2.30 suggests that the small-signal resistance looking into the base terminal is r_π. Since V_{CC} is replaced with a small-signal ground in a small-signal analysis, R_1 is parallel to R_2 and r_π. R_{in} is therefore $R_{in} = R_1||R_2||r_\pi$. Note that $R_{in} \approx r_\pi$ if $R_1 >> r_\pi$ and $R_2 >> r_\pi$.

Looking into the collector terminal, the small-signal resistance is r_o per Eq.2.51 since R_E is bypassed by C_E. Since V_{CC} is replaced with a small-signal ground in a small-signal analysis, $R_{out} = r_o||R_C$. $R_{out} \approx R_C$ if $r_o >> R_C$,

Common-Base Amplifier

V_{CC} is replaced by a small-signal ground in a small-signal analysis. R_1 is in parallel with R_2 and C_B. Because C_B is a bypass capacitor, the parallel combination of R_1 and R_2 is bypassed and the base terminal is grounded at small-signal frequencies. The small-signal resistance

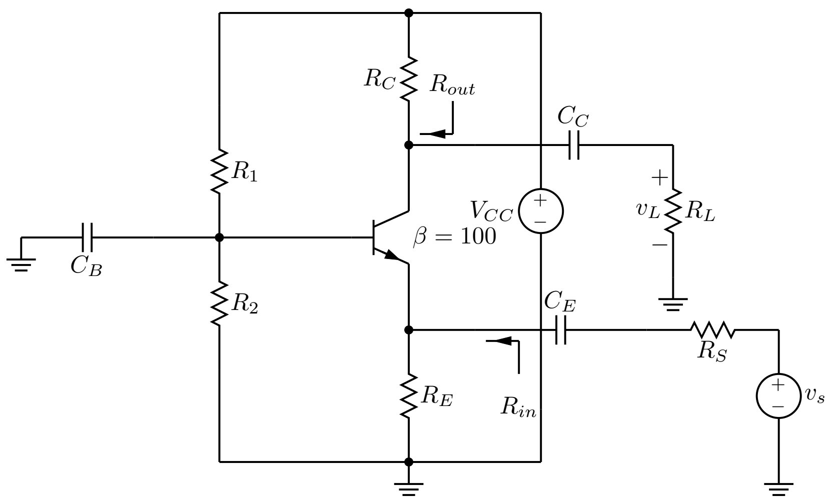

Figure 2.12: Input and output resistance of a common-base amplifier.

looking into the emitter terminal is $1/g_m$ since $R_B = 0$ in Eq. 2.30. $R_{in} = R_E \parallel \dfrac{1}{g_m}$. $R_{in} \approx \dfrac{1}{g_m}$ if $R_E \gg \dfrac{1}{g_m}$.

The resistance looking into the collector terminal is given by Eq. 2.51. R_E in Eq.2.51 is replaced by the parallel combination of R_E and R_S since C_E is a short and v_s is set to zero in a Thevenin equivalent resistance calculation.

$$R_{out} = R_C \parallel (r_o + (1 + g_m r_o)(R_E \parallel R_S \parallel r_\pi))\,.$$

$R_{out} \approx R_C$, if $r_o = \infty$.

Common-Collector Amplifier

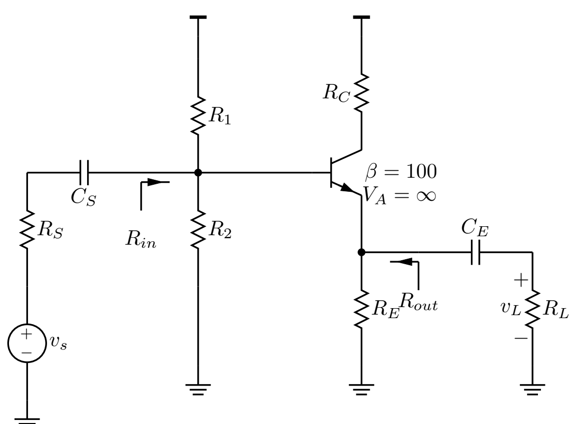

Figure 2.13: Input and output resistance of a common-collector amplifier.

The small-signal resistance looking into the base is given by Eq. 2.30 with R_E replaced by $R_E || R_L$ because C_E acts as a short. V_{CC} is replaced by a small-signal ground in a small-signal analysis. R_{in} is the parallel combination of R_1, R_2, and the small-signal resistance looking into the base, i.e. $R_{in} = R_1||R_2||(r_\pi + (\beta+1)R_E||R_L)$. $R_{in} \approx \beta R_E||R_L$ if $R_E||R_L >> \frac{1}{g_m}$ and R_1 and R_2 are greater than $r_\pi + (\beta + 1)R_E||R_L$.

With $V_A = \infty$, r_o is ∞. The small-signal resistance looking into the emitter terminal is given by Eq. 2.41 with C_S replaced by a short and v_s set to zero in a Thevenin resistance calculation, i.e. $\frac{R_1||R_2||R_S}{\beta+1} + \frac{1}{g_m}$. Assuming that R_S is much less than R_1 and R_2, the resistance looking into the emitter is approximately $\frac{R_S}{\beta+1} + \frac{1}{g_m}$, or simply $\frac{1}{g_m}$ if $\frac{R_S}{\beta+1} << \frac{1}{g_m}$. Thus, $R_{out} = R_E||\frac{1}{g_m}$. With $R_E >> \frac{1}{g_m}$, $R_{out} \approx \frac{1}{g_m}$.

2.7 *Voltage Gains*

2.7.1 *Common-Emitter Amplifier*

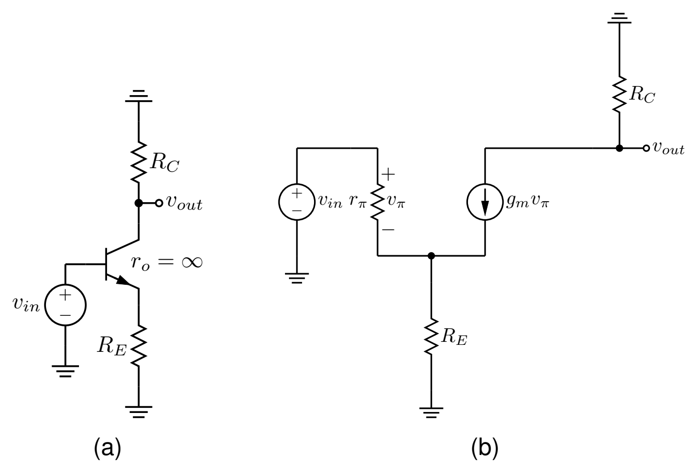

Figure 2.14: Common-emitter amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

Figure 2.14 is used to derive the small-signal gain of a common-emitter amplifier for $r_o = \infty$. The goal is to show that

$$A_V = \frac{v_{\text{out}}}{v_{\text{in}}} \approx -\frac{R_C}{\frac{1}{g_m} + R_E}.$$

From the small-signal collector circuit, the controlled current source is

$$g_m v_\pi.$$

Since this current flows through R_C, the output voltage is

$$v_{\text{out}} = -g_m v_\pi R_C. \tag{2.53}$$

Applying KCL at the emitter node gives

$$i_{R_E} = \frac{v_\pi}{r_\pi} + g_m v_\pi.$$

Therefore, the small-signal voltage across the emitter resistor is

$$v_{R_E} = i_{R_E} R_E = \left(\frac{v_\pi}{r_\pi} + g_m v_\pi \right) R_E. \tag{2.54}$$

From the input loop,

$$v_\pi = v_{\text{in}} - v_{R_E}.$$

Substituting (2.54) into this relation,

$$v_\pi = v_{\text{in}} - \left(\frac{v_\pi}{r_\pi} + g_m v_\pi \right) R_E.$$

Collecting the v_π terms,

$$v_\pi \left(1 + \frac{R_E}{r_\pi} + g_m R_E \right) = v_{\text{in}}. \tag{2.55}$$

Hence,

$$v_\pi = \frac{v_{\text{in}}}{1 + \frac{R_E}{r_\pi} + g_m R_E}. \tag{2.56}$$

Substitute (2.56) into (2.53):

$$v_{\text{out}} = -g_m R_C \frac{v_{\text{in}}}{1 + \frac{R_E}{r_\pi} + g_m R_E}.$$

Therefore,

$$A_V = \frac{v_{\text{out}}}{v_{\text{in}}} = -\frac{g_m R_C}{1 + \frac{R_E}{r_\pi} + g_m R_E}. \tag{2.57}$$

Dividing numerator and denominator by g_m gives

$$A_V = -\frac{R_C}{\frac{1}{g_m} + \frac{R_E}{g_m r_\pi} + R_E}.$$

Using

$$r_\pi = \frac{\beta}{g_m},$$

we obtain

$$\frac{R_E}{g_m r_\pi} = \frac{R_E}{\beta}.$$

For large β, the term R_E/β is small and can be neglected. Thus,

$$A_V \approx -\frac{R_C}{\frac{1}{g_m} + R_E}. \tag{2.58}$$

In conclusion, with $r_o = \infty$ and neglecting the small R_E/β term,

$$\boxed{A_V \approx -\frac{R_C}{\frac{1}{g_m} + R_E}}.$$

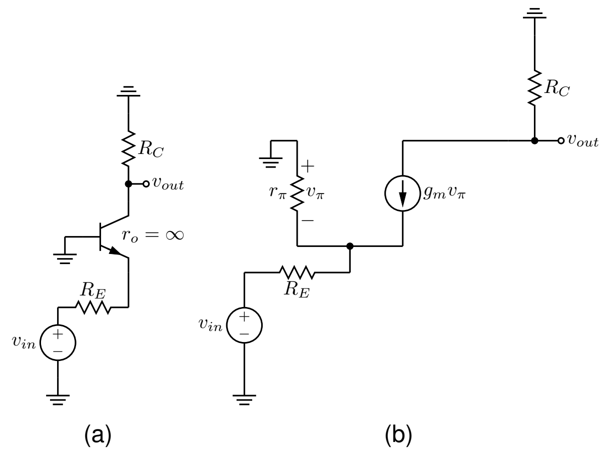

Figure 2.15: Common-base amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

2.7.2 *Common-Base Amplifier*

The small-signal voltage at the collector terminal is,

$$v_{\text{out}} = -g_m v_\pi R_C.$$

Therefore,

$$\frac{v_{\text{out}}}{v_{\text{in}}} = -g_m R_C \frac{v_\pi}{v_{\text{in}}}.$$

Next, use KCL at the emitter node.

$$\frac{v_\pi}{r_\pi} + g_m v_\pi = i_{R_E}.$$

Since the current through R_E is set by the voltage across it,

$$i_{R_E} = \frac{-v_\pi - v_{\text{in}}}{R_E}.$$

Thus,

$$\frac{v_\pi}{r_\pi} + g_m v_\pi = \frac{-v_\pi - v_{\text{in}}}{R_E}.$$

Multiplying by R_E gives

$$\left(\frac{v_\pi}{r_\pi} + g_m v_\pi\right) R_E = -v_\pi - v_{\text{in}}.$$

Rearrange:

$$-v_\pi - \frac{v_\pi}{r_\pi} R_E - g_m v_\pi R_E = v_{\text{in}}.$$

Factor out $-v_\pi$:

$$-v_\pi \left(1 + \frac{R_E}{r_\pi} + g_m R_E\right) = v_{\text{in}}.$$

So,

$$-v_\pi = \frac{v_{\text{in}}}{1 + \frac{R_E}{r_\pi} + g_m R_E}.$$

Using the large-β approximation, $R_E/r_\pi << g_m R_E$, so that

$$-v_\pi \approx \frac{v_{\text{in}}}{1 + g_m R_E}.$$

Hence,

$$v_\pi \approx -\frac{v_{\text{in}}}{1 + g_m R_E}.$$

Substitute into the output-voltage relation:

$$v_{\text{out}} = -g_m R_C \left(-\frac{v_{\text{in}}}{1 + g_m R_E} \right) = \frac{g_m R_C}{1 + g_m R_E} v_{\text{in}}.$$

Therefore,

$$A_V = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{g_m R_C}{1 + g_m R_E}.$$

Divide numerator and denominator by g_m:

$$A_V = \frac{R_C}{\frac{1}{g_m} + R_E}. \tag{2.59}$$

In conclusion, with $r_o = \infty$,

$$\boxed{A_V = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{R_C}{\frac{1}{g_m} + R_E}.}$$

2.7.3 Common-Collector Amplifier

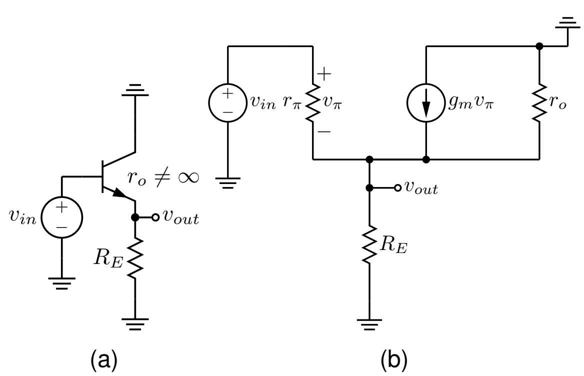

Figure 2.16: Common-collector amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

The small-signal voltage gain of the common-collector amplifier when $r_o \neq \infty$ is derived.

$$A_V = \frac{v_{out}}{v_{in}} = \frac{R_E \parallel r_o}{\frac{1}{g_m} + R_E \parallel r_o}.$$

Start with the definition of voltage gain:

$$A_V = \frac{v_{out}}{v_{in}}.$$

From the small-signal model,

$$v_\pi = v_{in} - v_{out}.$$

Apply KCL at the emitter node (which is also the output node). The currents entering from the transistor are

$$g_m v_\pi \qquad \text{and} \qquad \frac{v_\pi}{r_\pi},$$

and the current leaving through the emitter-side resistance is

$$\frac{v_{out}}{R_E \parallel r_o}.$$

Therefore,

$$g_m v_\pi + \frac{v_\pi}{r_\pi} = \frac{v_{out}}{R_E \parallel r_o}.$$

Factor v_π:

$$v_\pi \left(g_m + \frac{1}{r_\pi} \right) = \frac{v_{out}}{R_E \parallel r_o}.$$

Substitute $v_\pi = v_{in} - v_{out}$:

$$(v_{in} - v_{out}) \left(g_m + \frac{1}{r_\pi} \right) = \frac{v_{out}}{R_E \parallel r_o}.$$

Expand and collect terms:

$$v_{in} \left(g_m + \frac{1}{r_\pi} \right) = v_{out} \left(g_m + \frac{1}{r_\pi} \right) + \frac{v_{out}}{R_E \parallel r_o}.$$

So,

$$v_{in} \left(g_m + \frac{1}{r_\pi} \right) = v_{out} \left(g_m + \frac{1}{r_\pi} + \frac{1}{R_E \parallel r_o} \right).$$

Divide both sides by v_{in} and by the coefficient of v_{out}:

$$A_V = \frac{v_{out}}{v_{in}} = \frac{g_m + \dfrac{1}{r_\pi}}{g_m + \dfrac{1}{r_\pi} + \dfrac{1}{R_E \parallel r_o}}.$$

Multiply numerator and denominator by $R_E \parallel r_o$:

$$A_V = \frac{\left(g_m + \dfrac{1}{r_\pi} \right) (R_E \parallel r_o)}{1 + \left(g_m + \dfrac{1}{r_\pi} \right) (R_E \parallel r_o)}.$$

For large β, the term $1/r_\pi$ is small compared with g_m, so

$$g_m + \frac{1}{r_\pi} \approx g_m.$$

Hence,

$$A_V \approx \frac{g_m (R_E \parallel r_o)}{1 + g_m (R_E \parallel r_o)}.$$

Finally, divide numerator and denominator by g_m:

$$A_V \approx \frac{R_E \parallel r_o}{\dfrac{1}{g_m} + R_E \parallel r_o}. \tag{2.60}$$

2.7.4 *Examples*

Common-Emitter Amplifier

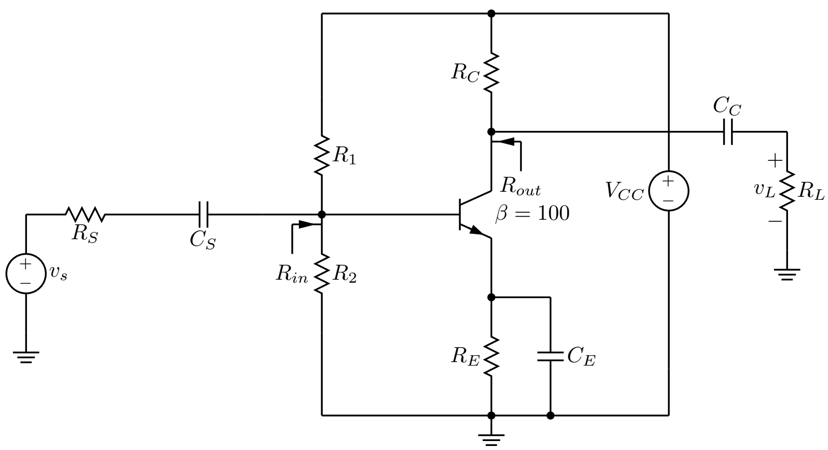

Figure 2.17: Small-signal voltage gain of a common-emitter amplifier.

The overall voltage gain from the source voltage v_s to the load voltage v_L is $\frac{v_L}{v_s}$. This gain can be written as

$$\frac{v_L}{v_s} = \frac{v_L}{v_b} \cdot \frac{v_b}{v_s},$$

where v_b is the small-signal voltage at the base of the transistor.

The source voltage v_s sees the source resistance R_S in series with the amplifier input resistance R_{in}. Therefore,

$$\frac{v_b}{v_s} = \frac{R_{in}}{R_{in} + R_S}.$$

This ratio tells us how much of the source voltage actually appears at the base.

The input resistance is the parallel combination of the bias resistors and the transistor input resistance:

$$R_{in} = R_1 \parallel R_2 \parallel R_b.$$

From Eq. 2.30, with R_E bypassed by C_E, we have

$$R_b = r_\pi.$$

Therefore,

$$R_{in} = R_1 \parallel R_2 \parallel r_\pi.$$

Substituting this expression into the voltage-divider equation gives

$$\frac{v_b}{v_s} = \frac{r_\pi \parallel R_1 \parallel R_2}{(r_\pi \parallel R_1 \parallel R_2) + R_S}.$$

Since the parallel combination is dominated by the smallest resistance, r_π,

$$r_\pi \parallel R_1 \parallel R_2 \approx r_\pi.$$

So the voltage-divider expression becomes

$$\frac{v_b}{v_s} \approx \frac{r_\pi}{r_\pi + R_S}.$$

If r_π is much larger than R_S, then

$$\frac{r_\pi}{r_\pi + R_S} \approx 1.$$

Thus,

$$\frac{v_b}{v_s} \approx 1.$$

This means that very little of the source voltage is lost at the input. In other words, the base voltage is approximately equal to the source voltage.

The gain from the base voltage to the load voltage is given by Eq. 2.58:

$$\frac{v_L}{v_b} = -g_m \left(R_C \parallel R_L\right).$$

Here, g_m is the transistor transconductance, and $R_C \parallel R_L$ is the effective load seen at the collector. The negative sign shows that the common-emitter amplifier inverts the signal. In this analysis, R_E is bypassed by C_E.

Now multiply the two gain factors together:

$$\frac{v_L}{v_s} = \frac{v_L}{v_b} \cdot \frac{v_b}{v_s}.$$

Substituting the expressions found above,

$$\frac{v_L}{v_s} = -g_m \left(R_C \parallel R_L\right) \frac{r_\pi \parallel R_1 \parallel R_2}{(r_\pi \parallel R_1 \parallel R_2) + R_S}.$$

Using the approximation

$$\frac{v_b}{v_s} \approx 1,$$

the overall gain simplifies to

$$\frac{v_L}{v_s} \approx -g_m \left(R_C \parallel R_L\right).$$

This result shows that if

$$R_1 \parallel R_2 \gg r_\pi \qquad \text{and} \qquad r_\pi \gg R_S,$$

then the input network causes very little attenuation. Under these conditions, the overall voltage gain is mainly determined by the transistor transconductance and the collector-side load:

$$A_v \approx -g_m \left(R_C \parallel R_L\right).$$

When the input attenuation is small, the common-emitter amplifier gain is approximately equal to the usual base-to-collector gain expression.

Common-Base Amplifier

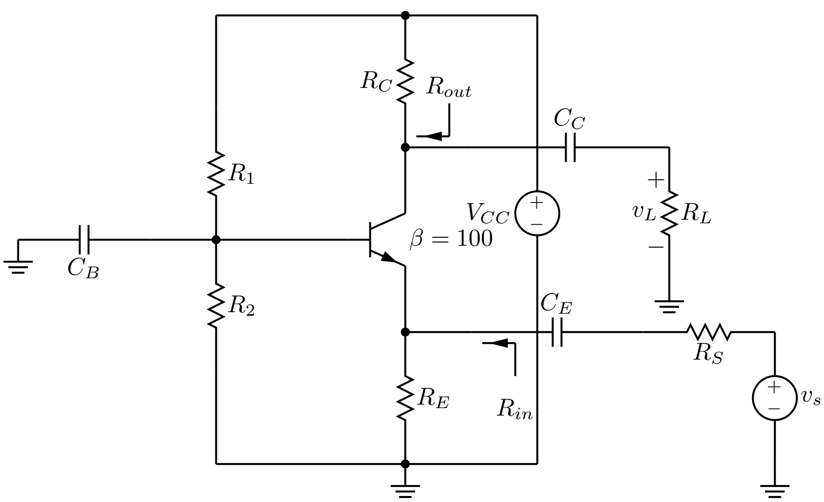

Figure 2.18: Small-signal voltage gain of a common-base amplifier.

The small-signal voltage gain of a common-base amplifier can be written as

$$\frac{v_L}{v_s} = \frac{v_e}{v_s} \cdot \frac{v_L}{v_e}.$$

This expression shows that the overall gain is the product of two factors: the fraction of the source voltage that appears at the emitter node, and the gain from the emitter node to the load voltage.

Because the source resistance R_S is in series with the input resistance R_{in}, only part of v_s appears at the emitter node. Therefore,

$$\frac{v_e}{v_s} = \frac{R_{in}}{R_{in} + R_S}.$$

The input resistance R_{in} is the parallel combination of R_e, the small-signal resistance seen looking into the emitter terminal, and R_E. According to Eq. 2.30,

$$R_e \approx \frac{1}{g_m},$$

because R_B is bypassed by C_B. Thus,

$$R_{in} = \frac{1}{g_m} \parallel R_E.$$

Substituting this result into the voltage-divider expression gives

$$\frac{v_e}{v_s} = \frac{\left(\frac{1}{g_m} \parallel R_E\right)}{\left(\frac{1}{g_m} \parallel R_E\right) + R_S}.$$

If R_{in} is designed to match R_S, then

$$\frac{1}{g_m} \parallel R_E = R_S.$$

Under this condition,

$$\frac{v_e}{v_s} = \frac{1}{2}.$$

In other words, one-half of the source voltage appears at the emitter.

According to Eq. 2.59, the gain from the emitter node to the load is

$$\frac{v_L}{v_e} = g_m \left(R_C \parallel R_L\right).$$

This means that once the small-signal voltage reaches the emitter, the circuit provides an additional gain factor of $g_m(R_C \parallel R_L)$ to produce the load voltage.

Combining these two results gives

$$\frac{v_L}{v_s} = \frac{\left(\frac{1}{g_m} \parallel R_E\right)}{\left(\frac{1}{g_m} \parallel R_E\right) + R_S} g_m \left(R_C \parallel R_L\right).$$

In the special case where

$$\frac{1}{g_m} \parallel R_E = R_S,$$

we have $v_e/v_s = 1/2$, so the overall gain simplifies to

$$\frac{v_L}{v_s} \approx \frac{g_m \left(R_C \parallel R_L\right)}{2}.$$

Common-Collector Amplifier

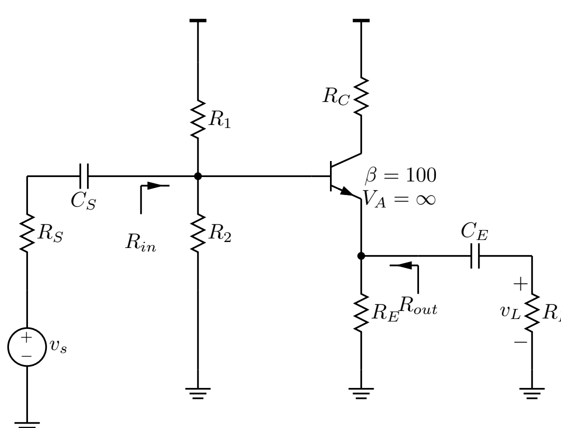

Figure 2.19: Small-signal gain of a common-collector amplifier.

The quantity $\frac{v_l}{v_s}$ is the small-signal voltage gain of a common-collector amplifier, also called an emitter follower. It shows how much of the source signal v_s appears at the output.

A convenient way to find the overall gain is to break it into two parts:

$$\frac{v_l}{v_s} = \frac{v_b}{v_s} \cdot \frac{v_l}{v_b}.$$

Here, $\frac{v_b}{v_s}$ describes how much of the source signal reaches the base, and $\frac{v_l}{v_b}$ describes the gain from the base to the emitter.

The input division from the source to the base is

$$\frac{v_b}{v_s} = \frac{R_{in}}{R_{in} + R_S}.$$

The input resistance is

$$R_{in} = R_1 \parallel R_2 \parallel R_b,$$

where, from Eq. 2.30,

$$R_b = r_\pi + (\beta + 1)(R_E \parallel R_L).$$

If $R_{in} \gg R_S$, then very little signal is lost across R_S, so

$$\frac{v_b}{v_s} \approx 1.$$

In other words, the base voltage is approximately equal to the source voltage.

Next, the gain from the base to the emitter is

$$\frac{v_l}{v_b} = \frac{R_E \parallel R_L}{\frac{1}{g_m} + R_E \parallel R_L}.$$

Multiplying these two factors gives the overall voltage gain:

$$\frac{v_l}{v_s} = \frac{v_b}{v_s} \cdot \frac{v_l}{v_b} = \frac{R_1 \parallel R_2 \parallel R_b}{R_1 \parallel R_2 \parallel R_b + R_S} \cdot \frac{R_E \parallel R_L}{\frac{1}{g_m} + R_E \parallel R_L}.$$

If $R_{in} \gg R_S$, then $v_b \approx v_s$, and the gain simplifies to

$$\frac{v_l}{v_s} \approx \frac{R_E \parallel R_L}{\frac{1}{g_m} + R_E \parallel R_L}.$$

Finally, if $R_E \parallel R_L \gg 1/g_m$, then the denominator is dominated by $R_E \parallel R_L$. Therefore,

$$\frac{v_l}{v_s} \approx 1.$$

This result shows why the circuit is called an emitter follower: the emitter voltage closely follows the input signal.

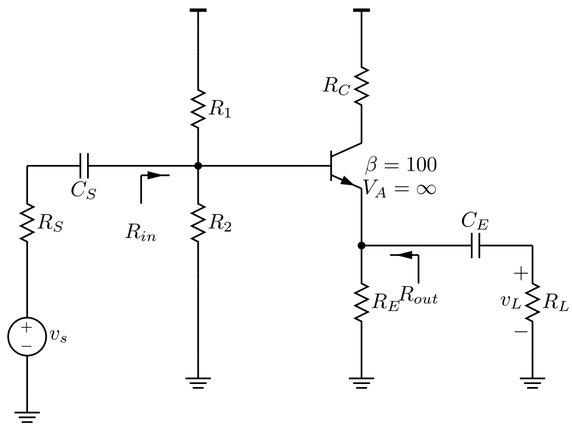

Figure 2.20: Design of a common-collector amplifier.

2.8 Design

2.8.1 Design of a Common-Collector Amplifier

A common-collector amplifier in Fig. 2.20 is designed to meet the following specifications:

- $\frac{v_L}{v_s} = \frac{19}{20}$ V/V
- $R_E \gg R_L$, i.e. $R_E = 10R_L$
- The load resistance R_L is given.
- $I_{R1} = 10\frac{I_C}{\beta}$

For a common-collector stage, the overall gain can be written as

$$\frac{v_L}{v_s} \approx \frac{v_L}{v_b} \qquad \text{if } R_{in} \gg R_S$$

This means that if the amplifier input resistance is much larger than the source resistance, then very little input signal is lost at the source, so the overall gain is approximately the same as the base-to-output gain.

For an emitter follower, the small-signal gain from base to emitter is

$$\frac{v_L}{v_b} \approx \frac{R_E \parallel R_L}{\frac{1}{g_m} + (R_E \parallel R_L)}$$

If $R_E \gg R_L$,

$$R_E \parallel R_L \approx R_L$$

and the gain expression becomes

$$\frac{R_L}{\frac{1}{g_m} + R_L} = \frac{19}{20}$$

Now solve the previous equation step by step:

$$20R_L = 19\left(\frac{1}{g_m} + R_L\right)$$

$$20R_L = \frac{19}{g_m} + 19R_L$$

$$R_L = \frac{19}{g_m}$$

Using the BJT small-signal relation

$$g_m = \frac{I_C}{V_T}$$

we substitute into

$$R_L = \frac{19}{g_m}$$

and get

$$R_L = \frac{19V_T}{I_C}$$

Therefore,

$$I_C = \frac{19V_T}{R_L}$$

This result gives the collector current needed to meet the gain specification once R_L is known.

After finding I_C, the note computes the emitter voltage as

$$V_E = I_C R_E$$

Then the base voltage is chosen as

$$V_B = V_E + V_{BE}$$

where V_{BE} is the usual base-emitter drop of the transistor.

Assume that the current through the divider is much larger than the base current. With

$$I_{R1} = 10I_B$$

the resistor values are then chosen from the DC voltages:

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

$$R_2 = \frac{V_B}{I_{R1}}$$

3
MOS

3.1 Small-Signal Resistance

3.1.1 Gate Resistance

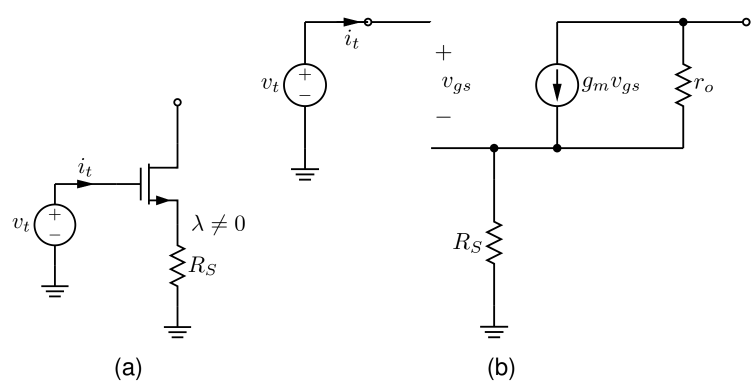

Figure 3.1: Gate resistance. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

The equivalent resistance seen looking into the gate terminal is found by applying a test voltage v_t at the gate and observing the resulting current, i_t, as shown in Fig. 3.1(a). When the transistor is replaced by its small-signal model, Fig. 3.1(b) shows that $i_t = 0$. Therefore, the equivalent resistance looking into the gate is infinite.

3.1.2 Source Resistance

The equivalent resistance into the source terminal is determined by applying a test voltage v_t at the source terminal and monitoring the resulting current (i_t) as shown in Fig. 3.2(a).

$$R_{eq} = \frac{v_t}{i_t}.$$

Note the followings from Fig. 3.2(b):

- the gate is at ac ground, so

$$v_{gs} = v_g - v_s = 0 - v_t = -v_t;$$

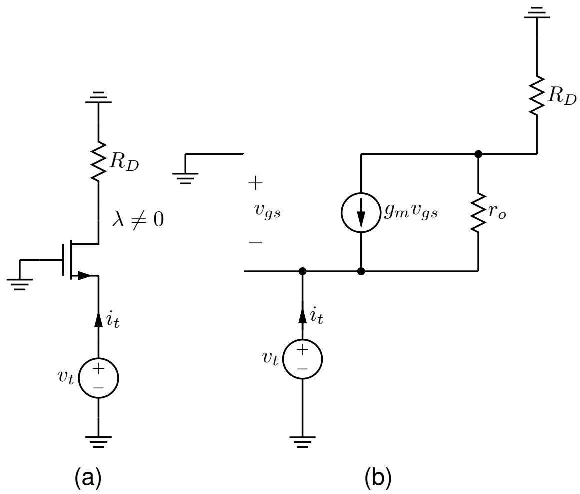

Figure 3.2: Source resistance. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its s mall-signal model.

- the drain voltage is set by the current through R_D, so

$$v_d = i_t R_D;$$

- the current through r_o is

$$\frac{v_t - v_d}{r_o}.$$

Using the annotation, Kirchhoff's current law at the source node gives

$$i_t + g_m v_{gs} = \frac{v_t - v_d}{r_o}.$$

Substitute $v_{gs} = -v_t$ and $v_d = i_t R_D$:

$$i_t + g_m(-v_t) = \frac{v_t - i_t R_D}{r_o}.$$

Multiply by r_o:

$$i_t r_o - g_m r_o v_t = v_t - i_t R_D.$$

Collect the i_t terms on one side:

$$i_t r_o + i_t R_D = v_t + g_m r_o v_t.$$

Factor both sides:

$$i_t(r_o + R_D) = v_t(1 + g_m r_o).$$

Therefore,

$$\frac{v_t}{i_t} = \frac{r_o + R_D}{1 + g_m r_o}.$$

The resistance looking into the source terminal is

$$\boxed{R_{eq} = \frac{r_o + R_D}{1 + g_m r_o}}.$$

If $r_o \to \infty$, then

$$R_{eq} \approx \frac{1}{g_m}.$$

3.1.3 *Drain Resistance*

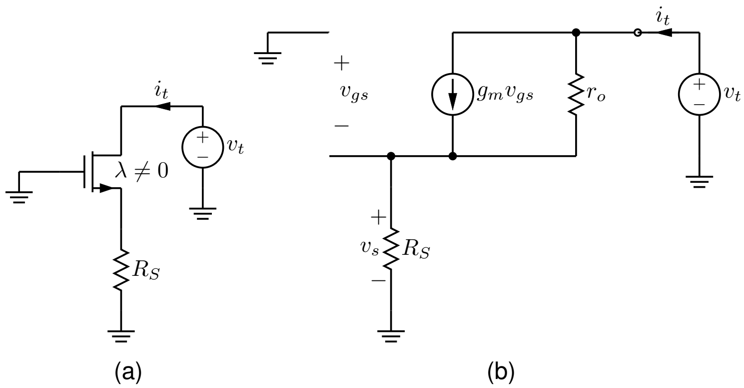

Figure 3.3: Drain resistance. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

The small-signal drain resistance is found by applying a test source, v_t, at the drain terminal and monitoring the resulting test current, as shown in Fig. 3.3(a). Thus,

$$R_{eq} = \frac{v_t}{i_t}.$$

Because the gate is at small-signal ground,

$$v_g = 0, \qquad v_{gs} = v_g - v_s = -v_s.$$

As shown in Fig. 3.3(b), the source resistor carries the test current, so

$$v_s = i_t R_S.$$

Therefore,

$$v_{gs} = -i_t R_S.$$

The current through r_o is

$$\frac{v_t - v_s}{r_o},$$

and the controlled current source contributes a current of

$$g_m v_{gs}.$$

Applying KCL at the drain gives

$$i_t = g_m v_{gs} + \frac{v_t - v_s}{r_o}.$$

Substituting $v_{gs} = -i_t R_S$ and $v_s = i_t R_S$ gives

$$i_t = g_m(-i_t R_S) + \frac{v_t - i_t R_S}{r_o}.$$

Multiplying both sides by r_o yields

$$i_t r_o = -g_m r_o i_t R_S + v_t - i_t R_S.$$

Rearranging to isolate v_t gives

$$v_t = i_t r_o + g_m r_o i_t R_S + i_t R_S.$$

Factoring out i_t gives

$$v_t = i_t \left(r_o + (1 + g_m r_o) R_S \right).$$

Therefore, the resistance seen by the test source is

$$R_{eq} = \frac{v_t}{i_t} = r_o + (1 + g_m r_o) R_S.$$

$$\boxed{R_{eq} = r_o + (1 + g_m r_o) R_S}$$

3.1.4 *Effective Resistance Between Gate and Drain*

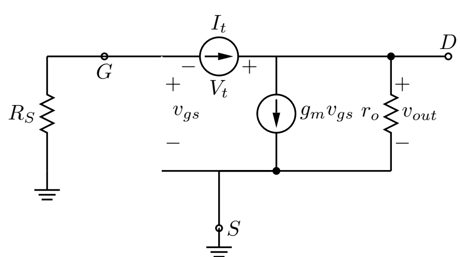

Figure 3.4: Effective resistance between gate and drain.

The goal is to find the equivalent resistance seen by the capacitor connected between the gate and drain. Denote this resistance by

$$R_{GD} = R_\mu = \frac{V_t}{I_t}.$$

A test source V_t is applied between the gate and drain, and the corresponding current is I_t.

Voltages used in the derivation With the source terminal grounded, the small-signal source voltage is

$$v_s = 0.$$

Therefore,

$$v_{gs} = v_g - v_s = v_g.$$

The gate node is connected to the source resistance R_S to ground, so the marked work gives

$$v_g = -I_t R_S.$$

At the drain node, the small-signal voltage is the output voltage,

$$v_d = v_{out}.$$

Because the test source is connected from gate to drain,

$$V_t = v_d - v_g.$$

Substituting $v_g = -I_t R_S$ gives

$$V_t = v_d + I_t R_S.$$

KCL at the drain node At the drain node, the test current I_t splits into the current through r_o and the dependent current source. Thus,

$$I_t = \frac{v_d}{r_o} + g_m v_{gs}.$$

Since $v_{gs} = v_g = -I_t R_S$,

$$I_t = \frac{v_d}{r_o} + g_m(-I_t R_S).$$

Hence,

$$I_t = \frac{v_d}{r_o} - g_m I_t R_S.$$

Move the second term to the left side:

$$I_t + g_m I_t R_S = \frac{v_d}{r_o}.$$

Factor I_t:

$$I_t(1 + g_m R_S) = \frac{v_d}{r_o}.$$

Multiply by r_o:

$$v_d = I_t\, r_o(1 + g_m R_S).$$

Expanding the product,

$$v_d = I_t\big(r_o + g_m r_o R_S\big).$$

Compute the test resistance Now substitute this expression for v_d into

$$V_t = v_d + I_t R_S.$$

Then

$$V_t = I_t\big(r_o + g_m r_o R_S\big) + I_t R_S.$$

Factor out I_t:

$$V_t = I_t\big(r_o + R_S + g_m r_o R_S\big).$$

Rearranging the terms,

$$V_t = I_t\Big(r_o + R_S(1 + g_m r_o)\Big).$$

Therefore,

$$R_{GD} = \frac{V_t}{I_t} = r_o + R_S(1 + g_m r_o).$$

Final result The equivalent resistance seen by C_{GD} (or C_μ) is

$$\boxed{R_{GD} = R_\mu = r_o + R_S(1 + g_m r_o).} \tag{3.1}$$

This matches the logic of the marked derivation: first express v_g in terms of the test current, then use KCL at the drain node, and finally compute V_t/I_t.

3.1.5 *Examples*

3.1.5.1 *Common-Gate Amplifier*

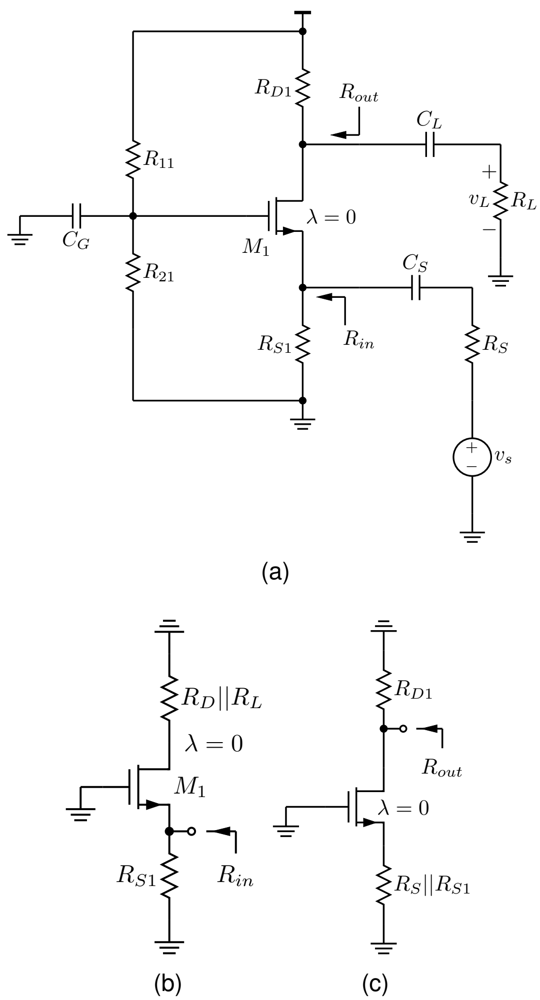

Figure 3.5: Common-gate amplifier. (a) Complete schematic. (b) Small-signal circuit used to calculate R_{in}. (c) Small-signal circuit used to calculate R_{out}.

For midband small-signal analysis, the capacitors C_G, C_S, and C_L are treated as shorts. The gate is at small-signal ground. Since $\lambda = 0$, the transistor output resistance is taken to be infinite, so $r_o = \infty$. v_s is set to zero when calculating for for small-signal resistances.

Input Resistance From Fig. 3.5(b), the small-signal resistance looking into the source of the transistor, with the gate at AC ground, is approximately $1/g_m$. Because R_{S1} is connected from the same node to ground, the total input resistance is

$$R_{in} = R_{S1} \parallel \frac{1}{g_m}.$$

Output Resistance To find R_{out}, v_s is set to zero as shown in Fig. 3.5(c). Since $\lambda = 0$, the transistor has $r_o = \infty$, so it does not provide a finite small-signal path from drain to source. Therefore, the only resistance seen at the drain is

$$R_{out} = R_{D1}.$$

3.1.5.2 *Common-Drain Amplifier*

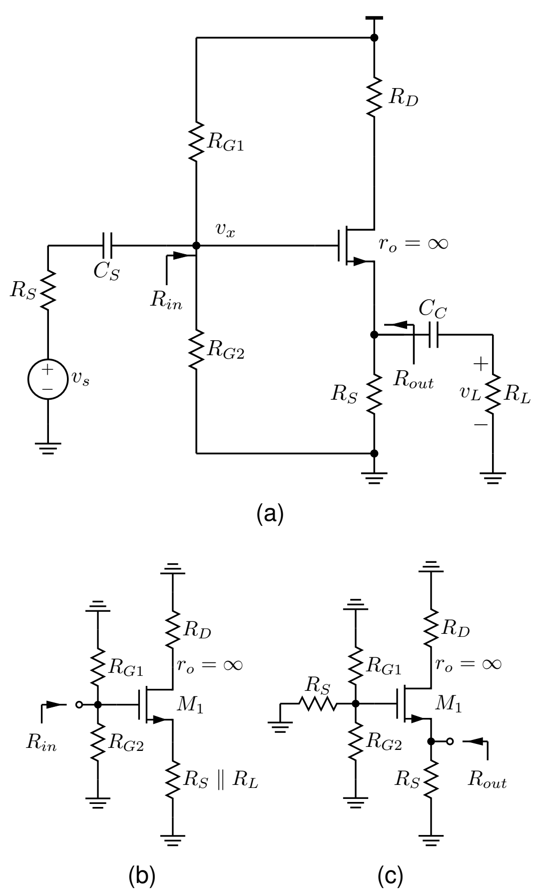

Figure 3.6: Common-drain amplifier. (a) Complete schematic. (b) Small-signal circuit used to calculate R_{in}. (c) Small-signal circuit used to calculate R_{out}.

For DC, the coupling capacitors are open circuits. For midband small-signal analysis, the coupling capacitors C_S and C_C are treated as short circuits, and the supply V_{DD} is taken to be AC ground. The problem also states that

$$r_o = \infty,$$

so the transistor output resistance is neglected.

Under these assumptions, the gate node is driven through R_S, the gate is shunted to ground by the bias network R_{G1} and R_{G2}, and the source node drives the load through the output coupling capacitor. Since C_C is a short at midband, the load resistor appears directly at the source node. Therefore the source sees

$$R_S \parallel R_L.$$

Input Resistance Because the MOSFET gate draws negligible small-signal current, the input resistance (Fig. 3.6(b)) is set by the gate-bias resistors only. Therefore,

$$R_{in} = R_{G1} \parallel R_{G2}.$$

Output resistance To find R_{out}, set the input source to zero and look back into the source terminal as shown in Fig. 3.6(c). With the input source suppressed, the gate is at AC ground. Since $r_o = \infty$, the resistance seen looking into the source of the transistor is

$$\frac{1}{g_m}.$$

This appears in parallel with the physical source resistor, so the output resistance is

$$R_{out} = R_S \parallel \frac{1}{g_m}.$$

3.2 *Voltage Gains*

3.2.1 *Common-Source Amplifier*

This section presents a derivation voltage gain of a common-source amplifier with source degeneration in Fig. 3.7. The small-signal model assumes $r_o = \infty$, so the transistor output resistance is neglected.

Goal Determine the voltage gain

$$A_v = \frac{v_{out}}{v_{in}}.$$

Step 1: Output relation In the small-signal model, the drain current is

$$i_d = g_m v_{gs}.$$

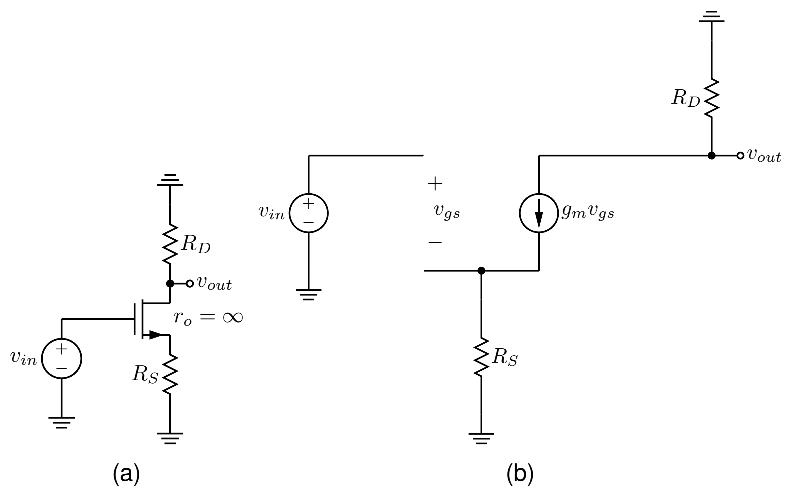

Figure 3.7: Common-source amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

Since the drain resistor R_D is connected to AC ground, the output voltage is

$$v_{out} = -i_d R_D = -g_m v_{gs} R_D.$$

Therefore,

$$v_{out} = -g_m R_D \, v_{gs}.$$

Step 2: Express v_{gs} in terms of v_{in} The gate is driven directly by the source v_{in}, so

$$v_g = v_{in}.$$

The source resistor carries the small-signal current produced by the controlled source. Thus the source voltage is

$$v_s = g_m v_{gs} R_S.$$

Using the definition of v_{gs},

$$v_{gs} = v_g - v_s = v_{in} - g_m v_{gs} R_S.$$

Rearranging gives

$$v_{gs} + g_m R_S v_{gs} = v_{in},$$

or

$$v_{gs}(1 + g_m R_S) = v_{in}.$$

Hence,

$$v_{gs} = \frac{v_{in}}{1 + g_m R_S}.$$

Step 3: Substitute into the output equation Substitute this result into the expression for v_{out}:

$$v_{out} = -g_m R_D \left(\frac{v_{in}}{1 + g_m R_S} \right).$$

So the gain is

$$\frac{v_{out}}{v_{in}} = -\frac{g_m R_D}{1 + g_m R_S}.$$

Divide the numerator and denominator by g_m to obtain the form emphasized in the annotation:

$$\boxed{\frac{v_{out}}{v_{in}} = -\frac{R_D}{1/g_m + R_S}.}$$

Final result For a common-source amplifier with source degeneration R_S and with $r_o = \infty$,

$$\boxed{A_v = \frac{v_{out}}{v_{in}} = -\frac{R_D}{1/g_m + R_S}.}$$

This result shows that the source resistor reduces the magnitude of the gain by introducing local negative feedback.

3.2.2 *Common-Gate Amplifier*

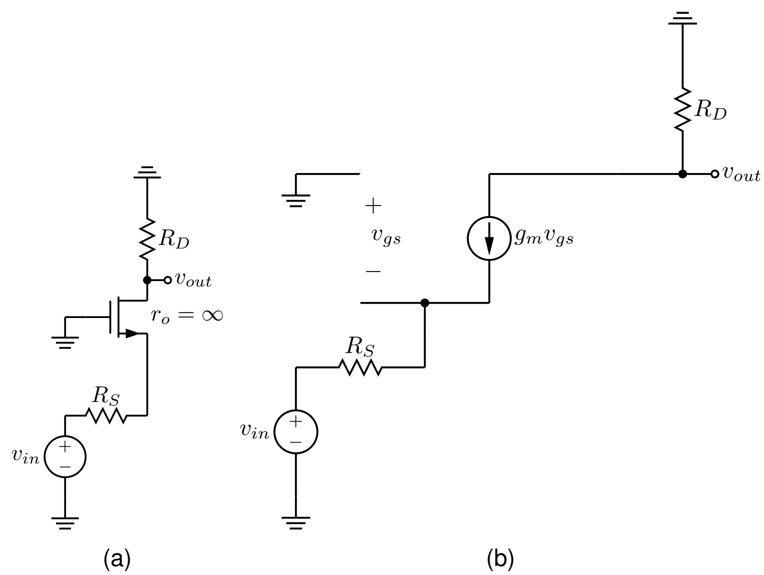

Figure 3.8: Common-gate amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

This section presents the voltage-gain derivation of a common-gate amplifier shown in Fig. 3.8. The transistor is analyzed with $r_o = \infty$, the gate is at ac ground, the input is applied at the source through R_S, and the output is taken at the drain through R_D.

1. Define the voltage gain The voltage gain is defined as

$$A_V = \frac{v_{out}}{v_{in}}.$$

Because the gate is connected to ac ground,

$$v_g = 0.$$

Therefore,

$$v_{gs} = v_g - v_s = 0 - v_s = -v_s.$$

This is the first key observation in the marked derivation: the small-signal gate-to-source voltage is the negative of the source voltage.

2. *Express the output voltage* With $r_o = \infty$, the small-signal drain current source is $g_m v_{gs}$, and the drain resistor R_D converts that current into the output voltage. Hence,

$$v_{out} = 0 - g_m v_{gs} R_D = -g_m v_{gs} R_D.$$

Since $v_{gs} = -v_s$,

$$v_{out} = -g_m(-v_s)R_D = g_m R_D \, v_s.$$

So once v_s is related to v_{in}, the gain follows immediately.

3. *Relate v_s to v_{in}* The annotation next writes the voltage across R_S in terms of the controlled current source. The current through R_S is the same current set by the transistor model, namely $g_m v_{gs}$. Therefore the drop across R_S is

$$(g_m v_{gs})R_S.$$

Writing a source-node relation gives

$$v_s = v_{in} + g_m v_{gs} R_S.$$

Now substitute $v_{gs} = -v_s$:

$$v_s = v_{in} + g_m(-v_s)R_S,$$

so

$$v_s = v_{in} - g_m R_S v_s.$$

Move all terms in v_s to one side:

$$v_s(1 + g_m R_S) = v_{in}.$$

Thus,

$$v_s = \frac{v_{in}}{1 + g_m R_S}.$$

This matches the boxed result in the annotation.

4. *Substitute into the output expression* Substituting the expression for v_s into the result for v_{out} gives

$$v_{out} = g_m R_D \left(\frac{v_{in}}{1 + g_m R_S} \right).$$

Therefore,

$$A_V = \frac{v_{out}}{v_{in}} = \frac{g_m R_D}{1 + g_m R_S}.$$

Dividing numerator and denominator by g_m gives the equivalent and often more useful form

$$A_V = \frac{R_D}{1/g_m + R_S}.$$

Final result For the common-gate amplifier with $r_o = \infty$,

$$\boxed{\frac{v_{out}}{v_{in}} = \frac{g_m R_D}{1 + g_m R_S} = \frac{R_D}{1/g_m + R_S}}.$$

The gain is positive, which is consistent with the common-gate amplifier: increasing the source input produces an output at the drain with the same small-signal polarity.

3.2.3 Common-Drain Amplifier

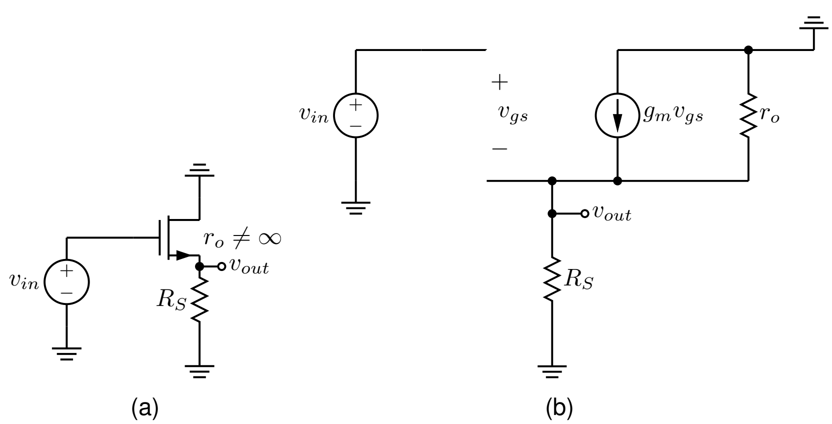

Figure 3.9: Common-drain amplifier. (a) Small-signal circuit. (b) Small-signal circuit with the transistor replaced by its small-signal model.

This section presents the voltage gain derivation of the common-drain amplifier (a.k.a. the source follower) shown in Fig. 3.9.

Goal We want the voltage gain

$$A_V = \frac{v_{out}}{v_{in}}.$$

The circuit has a source resistor R_S and includes the transistor output resistance r_o, so the small-signal resistance from the source node to ground is

$$R_S \parallel r_o.$$

Step 1: Write v_{gs} in terms of v_{in} and v_{out} For a common-drain stage, the gate is the input and the source is the output. Therefore,

$$v_{gs} = v_g - v_s.$$

Since $v_g = v_{in}$ and $v_s = v_{out}$,

$$v_{gs} = v_{in} - v_{out}.$$

Step 2: Write the output voltage from the small-signal model In the small-signal model, the controlled current source has value $g_m v_{gs}$, and that current flows through the equivalent resistance seen at the output node, namely $R_S \parallel r_o$. Hence,

$$v_{out} = g_m v_{gs}(R_S \parallel r_o).$$

Substitute $v_{gs} = v_{in} - v_{out}$:

$$v_{out} = g_m(R_S \parallel r_o)(v_{in} - v_{out}).$$

Step 3: Solve for the gain Expand the previous equation:

$$v_{out} = g_m(R_S \parallel r_o)v_{in} - g_m(R_S \parallel r_o)v_{out}.$$

Move the v_{out} terms to one side:

$$v_{out}\big(1 + g_m(R_S \parallel r_o)\big) = g_m(R_S \parallel r_o)v_{in}.$$

Now divide by v_{in}:

$$\frac{v_{out}}{v_{in}} = \frac{g_m(R_S \parallel r_o)}{1 + g_m(R_S \parallel r_o)}.$$

This is the gain expression in the same form shown in the annotation.

Step 4: Rewrite the result Divide the numerator and denominator by g_m:

$$\frac{v_{out}}{v_{in}} = \frac{R_S \parallel r_o}{\dfrac{1}{g_m} + R_S \parallel r_o}.$$

Therefore,

$$A_V = \frac{R_S \parallel r_o}{\dfrac{1}{g_m} + R_S \parallel r_o}.$$

Special case: $r_o \to \infty$ If the transistor output resistance is very large, then

$$R_S \parallel r_o \approx R_S.$$

So the gain reduces to

$$A_V \approx \frac{R_S}{\dfrac{1}{g_m} + R_S}.$$

Final result The common-drain amplifier voltage gain is

$$\boxed{A_V = \frac{v_{out}}{v_{in}} = \frac{R_S \parallel r_o}{\dfrac{1}{g_m} + R_S \parallel r_o}}.$$

For $r_o \to \infty$, this becomes

$$\boxed{A_V \approx \frac{R_S}{\dfrac{1}{g_m} + R_S}}.$$

3.2.4 Examples

3.2.4.1 Common-Gate Amplifier

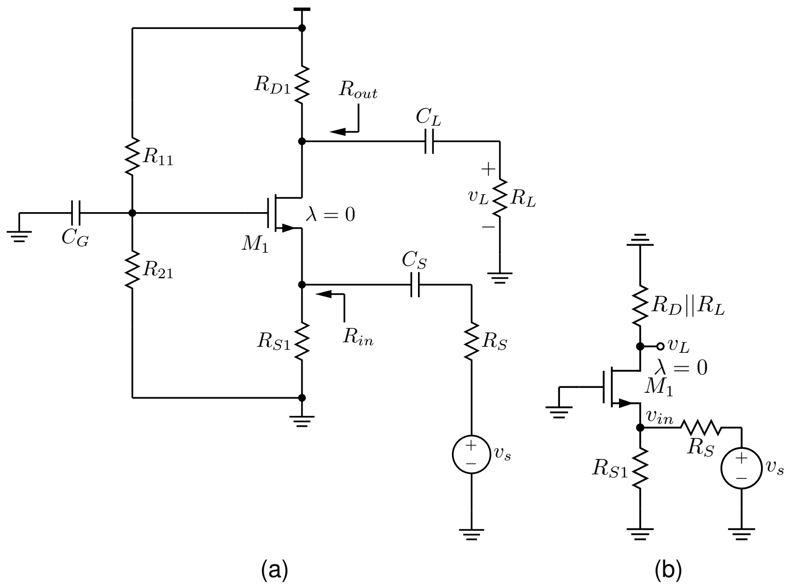

Figure 3.10: Common-gate amplifier. (a) Complete schematic. (b) Small-signal circuit.

For midband small-signal analysis, the capacitors C_G, C_S, and C_L are treated as shorts. Since $\lambda = 0$, the transistor output resistance is taken to be infinite, so $r_o = \infty$.

$\frac{v_L}{v_s}$ can be written as the product of the voltage division product $\frac{v_{in}}{v_s}$ and the amplifier gain $\frac{v_L}{v_{in}}$.

$$\frac{v_L}{v_s} = \frac{v_L}{v_{in}} \cdot \frac{v_{in}}{v_s}.$$

At the input, the signal source v_s drives the series resistor R_S and the amplifier input resistance R_{in}. Hence the input voltage is obtained by voltage division:

$$\frac{v_{in}}{v_s} = \frac{R_{in}}{R_S + R_{in}} = \frac{R_{S1} \parallel \frac{1}{g_m}}{R_S + R_{S1} \parallel \frac{1}{g_m}}.$$

Using the common-gate voltage gain equation, with R_D set to $R_{D1}||R_L$ and R_S set to 0,

$$\frac{v_L}{v_{in}} = g_m(R_{D1} \parallel R_L).$$

Combining $\frac{v_{in}}{v_s}$ and $\frac{v_L}{v_{in}}$ to give the overall voltage gain $\frac{v_L}{v_s}$:

$$\frac{v_L}{v_s} = g_m(R_{D1} \parallel R_L)\frac{R_{S1} \parallel \frac{1}{g_m}}{R_S + R_{S1} \parallel \frac{1}{g_m}}.$$

3.2.4.2 *Common Drain Amplifier*

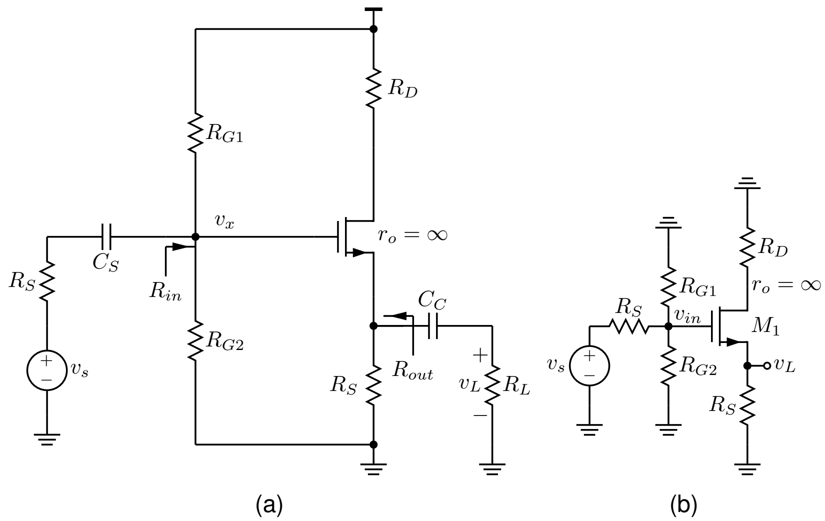

Figure 3.11: Common-drain amplifier. (a) Complete schematic. (b) Small-signal circuit.

The overall gain as

$$\frac{v_L}{v_s} = \frac{v_L}{v_{in}} \cdot \frac{v_{in}}{v_s},$$

where v_{in} is the gate voltage.

Since the gate current is negligible, the input side is a simple voltage divider formed by R_S and $R_{G1} \parallel R_{G2}$. Hence,

$$\frac{v_{in}}{v_s} = \frac{R_{G1} \parallel R_{G2}}{R_S + (R_{G1} \parallel R_{G2})}.$$

The second factor is the gain of the source follower from gate to source. Because the output is taken at the source and the source sees the resistance $R_S \parallel R_L$, the standard source-follower gain is

$$\frac{v_L}{v_{in}} = \frac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)}.$$

An equivalent form is

$$\frac{v_L}{v_{in}} = \frac{R_S \parallel R_L}{\frac{1}{g_m} + R_S \parallel R_L}.$$

Multiplying the two pieces gives

$$\frac{v_L}{v_s} = \left(\frac{R_{G1} \parallel R_{G2}}{R_S + (R_{G1} \parallel R_{G2})}\right)\left(\frac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)}\right).$$

Therefore,

$$\boxed{\frac{v_L}{v_s} = \left(\frac{R_{G1} \parallel R_{G2}}{R_S + (R_{G1} \parallel R_{G2})}\right)\left(\frac{R_S \parallel R_L}{\frac{1}{g_m} + R_S \parallel R_L}\right)}$$

which is the same result written in a simplified and organized form.

4
Multi-Stage Amplifier

4.1 CS-CC Amplifier

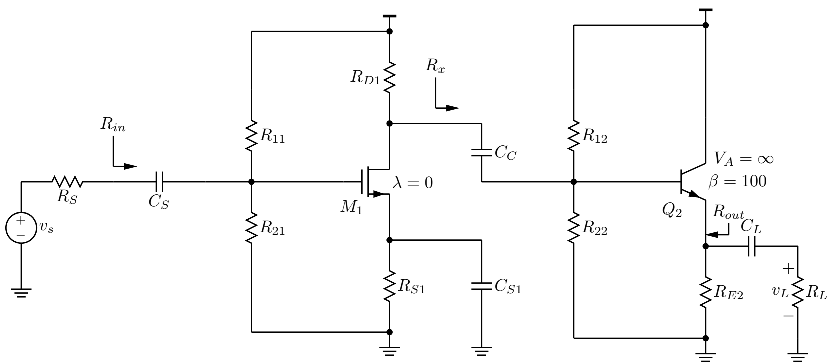

Figure 4.1: Schematic of a common-source common-collector amplifier.

4.1.1 Small-Signal Analysis

The overall voltage gain is decomposed into three factors:

$$\frac{v_l}{v_s} = \frac{v_l}{v_x} \cdot \frac{v_x}{v_y} \cdot \frac{v_y}{v_s}.$$

Here,

- v_y is the small-signal voltage at the gate terminal of M_1.
- v_x is the small-signal voltage at the base of Q_2.
- v_l is the load voltage.

4.1.2 Load stage: common-collector relation

From the small-signal emitter-follower model,

$$\frac{v_l}{v_x} = \frac{R_{E2} \parallel R_L}{\dfrac{1}{g_{m2}} + R_{E2} \parallel R_L}.$$

This is the voltage gain of the common-collector stage from its base node v_x to the load node v_l.

The output resistance of this stage is written in the notes as

$$R_{out} = R_{E2} \parallel R_{e2},$$

where the transistor looking into the emitter contributes the equivalent resistance

$$R_{e2} = \frac{R_{D1} \parallel R_{12} \parallel R_{22}}{\beta_2 + 1} + \frac{1}{g_{m2}}.$$

4.1.3 *Gain of the common-source stage*

The first stage is modeled as a common-source amplifier. The notes give

$$\frac{v_x}{v_y} \approx -g_{m1} \left(R_{D1} \parallel R_{b2} \parallel R_{12} \parallel R_{22} \right).$$

$R_{b2} = r_{\pi 2} + (\beta_2 + 1) R_{E2} \parallel R_L$. This is the voltage gain from the gate-controlled input node v_y to the drain node v_x, including the loading caused by the bias network and the input resistance of the next stage.

The input resistance seen by the source is

$$R_{in} = R_{11} \parallel R_{21},$$

so the source division factor is

$$\frac{v_y}{v_s} = \frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}}.$$

Combining the three pieces gives the complete small-signal relation:

$$\frac{v_l}{v_s} = \left(\frac{R_{E2} \parallel R_L}{\dfrac{1}{g_{m2}} + R_{E2} \parallel R_L} \right) \left[-g_{m1} \left(R_{D1} \parallel R_{b2} \parallel R_{12} \parallel R_{22} \right) \right] \left(\frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}} \right).$$

4.1.4 *Design Approximation*

The annotation then introduces a sequence of design approximations.

1. $R_{11} \parallel R_{22} \gg R_S$

   Then
   $$\frac{v_y}{v_s} = \frac{R_{11} \parallel R_{22}}{R_S + R_{11} \parallel R_{22}} \approx 1.$$
   So the source signal is transferred almost completely to node v_y.

2. $R_{E2} \gg R_{e2}$

   Then the output resistance of the common-collector stage becomes
   $$R_{out} = R_{E2} \parallel R_{e2} \approx R_{e2}.$$

3. $R_{12} \parallel R_{22} \gg R_{D1}$

   Then the gain of the first stage simplifies to

$$\frac{v_x}{v_y} \approx -g_{m1}\left(R_{D1} \parallel r_{\pi 2}\right).$$

   Also,

$$R_{e2} = \frac{1}{g_{m2}} + \frac{R_{D1} \parallel r_{\pi 2} \parallel R_{12} \parallel R_{22}}{\beta_2 + 1} \approx \frac{1}{g_{m2}} + \frac{R_{D1}}{\beta_2 + 1}.$$

4. $\dfrac{R_{D1}}{\beta_2 + 1} < \dfrac{1}{g_{m2}}$

   This further reduces the previous expressions to

$$\frac{v_x}{v_y} \approx -g_{m1} R_{D1},$$

   and

$$R_{out} \approx \frac{1}{g_{m2}} + \frac{R_{D1}}{\beta_2 + 1}.$$

   Under this condition,

$$R_{out} \approx \frac{1}{g_{m2}}.$$

5. $R_{E2} \gg R_L$

   Then the common-collector gain becomes

$$\frac{v_l}{v_x} = \frac{R_L}{\dfrac{1}{g_{m2}} + R_L} \approx \frac{R_L}{R_{out} + R_L}.$$

6. $R_{b2} \gg R_{D1}$

   The first-stage gain is approximated by

$$\frac{v_x}{v_y} \approx -g_{m1} R_{D1}.$$

4.1.5 *Simplified Design Equations*

With the above approximations, the overall gain reduces to

$$\frac{v_l}{v_s} \approx \frac{v_y}{v_s} \cdot \frac{v_x}{v_y} \cdot \frac{v_l}{v_x} = \left(\frac{R_{in}}{R_S + R_{in}}\right)\left(-g_{m1} R_{D1}\right)\left(\frac{R_L}{R_{out} + R_L}\right),$$

where

$$R_{in} = R_{11} \parallel R_{21}, \qquad R_{out} \approx \frac{1}{g_{m2}}.$$

Therefore the simplified design set is

$$\frac{v_y}{v_s} \approx \frac{R_{in}}{R_S + R_{in}}, \qquad \frac{v_x}{v_y} \approx -g_{m1} R_{D1}, \qquad \frac{v_l}{v_x} \approx \frac{R_L}{\dfrac{1}{g_{m2}} + R_L},$$

with

$$R_{in} = R_{11} \parallel R_{21}, \qquad R_{out} \approx \frac{1}{g_{m2}}.$$

4.1.6 *Design Procedure for the Common-Collector Amplifier*

The common-collector voltage gain is taken as

$$A_{V,CC} = \frac{v_l}{v_x} = \frac{R_L}{\dfrac{1}{g_{m2}} + R_L}.$$

Let $A_{V,CC}$ be

$$A_{V,CC} = \frac{19}{20}.$$

Substituting this target gives

$$\frac{19}{20} = \frac{R_L}{\dfrac{1}{g_{m2}} + R_L}.$$

Solving,

$$19\left(\frac{1}{g_{m2}} + R_L\right) = 20R_L \quad \Rightarrow \quad \frac{19}{g_{m2}} = R_L \quad \Rightarrow \quad g_{m2} = \frac{19}{R_L}.$$

Using the BJT relation

$$g_{m2} = \frac{I_{C2}}{V_T},$$

we obtain

$$I_{C2} = \frac{19V_T}{R_L}.$$

This is the collector current needed to obtain the chosen common-collector gain.

Let the emitter resistor to be much larger than the load, for example,

$$R_{E2} \approx 10R_L.$$

Once I_{C2} is known, the emitter voltage is approximated by

$$V_{E2} = I_{C2}R_{E2}.$$

Then the base voltage is

$$V_{B2} = V_{E2} + V_{BE}.$$

If the divider current through R_{22} is chosen as I_{R22}, the bias resistors are determined from

$$R_{22} = \frac{V_{B2}}{I_{R22}}, \qquad R_{12} = \frac{V_{CC} - V_{B2}}{I_{R22}}.$$

In summary, the common-collector design sequence is:

1. Choose the desired follower gain $A_{V,CC}$,
2. Solve for g_{m2},

3. Compute $I_{C2} = g_{m2} V_T$,
4. Choose R_{E2} so that $R_{E2} \gg R_L$,
5. Determine V_{E2} and then V_{B2},
6. Compute R_{12} and R_{22} from I_{R22}.

4.1.7 *Design Procedure for the Common-Source Amplifier*

The common-source stage is designed from the approximate gain relation

$$|A_{CS}| = g_{m1} R_{D1}.$$

The notes rewrite this as

$$|A_{CS}| = \frac{g_{m1}}{I_{D1}} (I_{D1} R_{D1}).$$

This form is useful because it separates the transistor choice, represented by g_{m1}/I_{D1}, from the resistor voltage drop $I_{D1} R_{D1}$.

Assume that g_{m1}/I_{D1}, V_{S1}, and $|A_{V,CS}|$ are chosen, e.g.,

1. $\dfrac{g_m}{I_D} \approx 10$ S/A,
2. $|A_V| = 40$ V/V,
3. $V_{S1} = 0.3$ V.

$$\frac{g_{m1}}{I_{D1}} = \frac{2}{V_{GS1} - V_t}.$$

Therefore,

$$V_{GS1} = \frac{2}{g_{m1}/I_{D1}} + V_t.$$

Once V_{GS1} is known, the drain current follows from the square-law expression

$$I_{D1} = K_n (V_{GS1} - V_t)^2.$$

The source resistor is obtained from the chosen source voltage:

$$R_{S1} = \frac{V_{S1}}{I_{D1}}.$$

The drain resistor follows from the gain equation:

$$R_{D1} = \frac{|A_{CS}|}{g_{m1}/I_{D1}} \cdot \frac{1}{I_{D1}}.$$

This is equivalent to solving $|A_{CS}| = g_{m1} R_{D1}$ after expressing g_{m1} as $(g_{m1}/I_{D1}) I_{D1}$.

The gate voltage is then set by

$$V_{GS1} = V_{G1} - V_{S1} \quad \Rightarrow \quad V_{G1} = V_{GS1} + V_{S1}.$$

Finally, if the divider current through R_{21} is selected as I_{R21}, the gate-bias resistors are

$$R_{21} = \frac{V_{G1}}{I_{R21}}, \qquad R_{11} = \frac{V_{CC} - V_{G1}}{I_{R21}}.$$

In conclusion, the common-source design sequence is:

1. Choose a target intrinsic efficiency g_m/I_D,
2. Choose the desired stage gain $|A_{CS}|$,
3. Choose V_{S1},
4. Compute V_{GS1} from g_m/I_D,
5. Compute I_{D1} from the device equation,
6. Find R_{S1} from V_{S1} and I_{D1},
7. Find R_{D1},
8. Compute $V_{G1} = V_{GS1} + V_{S1}$,
9. Compute R_{11} and R_{21}.

5
Frequency Response

5.1 Generic Frequency Response

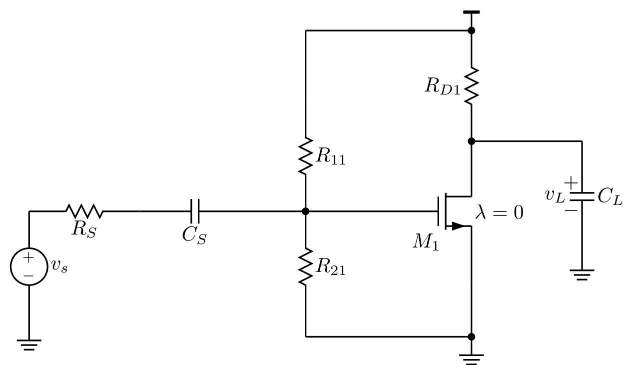

Figure 5.1: A generic common-source amplifier used to illustrate the low- and high-frequency behavior of the transfer function.

5.1.1 Frequency Response of a Common-Source Amplifier

In this section, we derive the transfer function of the common-source amplifier in Fig. 5.1 and express it in the standard form

$$H(\omega) = \frac{v_L}{v_s} = H_0 G(\omega).$$

The result separates naturally into a constant midband gain and two frequency-dependent factors: a low-frequency high-pass term produced by the input coupling capacitor and a high-frequency low-pass term produced by the load capacitor.

We will show that

$$G(\omega) = \frac{j\omega/\omega_L}{1 + j\omega/\omega_L} \cdot \frac{1}{1 + j\omega/\omega_H},$$

with

$$H_0 = -g_m R_{D1} \frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}},$$

$$\omega_L = \frac{1}{(R_S + R_{11} \parallel R_{21})C_S}, \qquad \omega_H = \frac{1}{R_{D1}C_L}.$$

Decomposition of the Gain

A convenient way to analyze the circuit is to write the overall gain as

$$\frac{v_L}{v_s} = \frac{v_L}{v_{in}} \cdot \frac{v_{in}}{v_s}.$$

This decomposition is useful because the factor v_L/v_{in} captures the drain-side behavior and reveals the high-frequency pole, while the factor v_{in}/v_s captures the input network and reveals the low-frequency pole.

Drain-Side Gain

With $\lambda = 0$, the transistor output resistance is infinite, and the small-signal drain current is

$$i_d = g_m v_{in}.$$

Because the output is taken at the drain, the voltage gain from the input node to the load voltage is

$$\frac{v_L}{v_{in}} = -g_m Z_D,$$

where Z_D is the impedance seen at the drain node.

At this node, both R_{D1} and C_L are connected to ac ground, so

$$Z_D = R_{D1} \parallel \frac{1}{j\omega C_L}.$$

Evaluating the parallel combination gives

$$Z_D = \frac{R_{D1}\left(\frac{1}{j\omega C_L}\right)}{R_{D1} + \frac{1}{j\omega C_L}} = \frac{R_{D1}}{1 + j\omega R_{D1} C_L}.$$

Therefore,

$$\frac{v_L}{v_{in}} = -g_m \frac{R_{D1}}{1 + j\omega R_{D1} C_L}.$$

Defining the high-frequency corner as

$$\omega_H = \frac{1}{R_{D1} C_L},$$

we obtain

$$\frac{v_L}{v_{in}} = -g_m R_{D1} \cdot \frac{1}{1 + j\omega/\omega_H}.$$

This is a first-order low-pass response with pole at ω_H.

Input Network Gain

Since the MOSFET gate draws negligible current, the resistance seen from the input node to ground is simply

$$R_G = R_{11} \parallel R_{21}.$$

Applying voltage division to the source resistance, coupling capacitor, and gate-bias network yields

$$\frac{v_{in}}{v_s} = \frac{R_G}{R_S + \dfrac{1}{j\omega C_S} + R_G}.$$

Multiplying the numerator and denominator by $j\omega C_S$ gives

$$\frac{v_{in}}{v_s} = \frac{j\omega C_S R_G}{1 + j\omega C_S(R_S + R_G)}.$$

Now factor the numerator so that the constant gain term appears explicitly:

$$\frac{v_{in}}{v_s} = \frac{j\omega C_S(R_S + R_G)}{1 + j\omega C_S(R_S + R_G)} \cdot \frac{R_G}{R_S + R_G}.$$

Defining the low-frequency corner as

$$\omega_L = \frac{1}{(R_S + R_G)C_S} = \frac{1}{(R_S + R_{11} \parallel R_{21})C_S},$$

we obtain

$$\frac{v_{in}}{v_s} = \frac{j\omega/\omega_L}{1 + j\omega/\omega_L} \cdot \frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}}.$$

This is a first-order high-pass response with corner frequency ω_L.

Overall Transfer Function

Multiplying the two factors gives

$$\frac{v_L}{v_s} = \frac{v_L}{v_{in}} \cdot \frac{v_{in}}{v_s}.$$

Substituting the expressions found above,

$$\frac{v_L}{v_s} = \left(-g_m R_{D1} \cdot \frac{1}{1 + j\omega/\omega_H}\right)\left(\frac{j\omega/\omega_L}{1 + j\omega/\omega_L} \cdot \frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}}\right).$$

Grouping the frequency-independent terms into a single constant produces the standard form

$$\frac{v_L}{v_s} = H_0 G(\omega),$$

where

$$H_0 = -g_m R_{D1} \frac{R_{11} \parallel R_{21}}{R_S + R_{11} \parallel R_{21}},$$

and

$$G(\omega) = \frac{j\omega/\omega_L}{1 + j\omega/\omega_L} \cdot \frac{1}{1 + j\omega/\omega_H}.$$

Interpretation of the Result

The transfer function contains three distinct parts:

1. the midband gain constant H_0,

2. a low-frequency high-pass factor associated with the coupling capacitor C_S,

3. a high-frequency low-pass factor associated with the load capacitor C_L.

For frequencies much smaller than ω_L, the coupling capacitor limits signal transmission and the gain is small. For frequencies in the range $\omega_L \ll \omega \ll \omega_H$, the reactive effects are negligible and the gain approaches the midband value H_0. For frequencies much larger than ω_H, the load capacitor shunts the output signal to ground, causing the gain to decrease again.

Thus, the circuit exhibits a band-pass type response with one lower corner frequency at ω_L and one upper corner frequency at ω_H.

5.1.1.1 Example

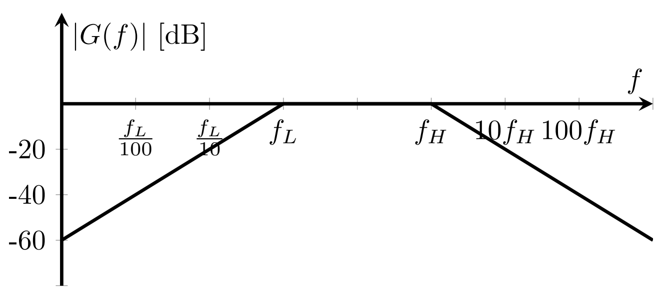

Figure 5.2: Magnitude response associated with $G(\omega)$.

This section examines the frequency-response of $G(\omega)$

$$G(\omega) = \frac{j\omega/\omega_L}{1 + j\omega/\omega_L} \cdot \frac{1}{1 + j\omega/\omega_H},$$

with

$$\omega_H = 100\omega_L.$$

The first factor is a high-pass term with corner frequency ω_L, and the second factor is a low-pass term with corner frequency ω_H. Together, these terms produce a band-pass response. To understand the behavior of $|G(\omega)|$, we evaluate the expression at three representative frequencies: one far below ω_L, one in the midband region, and one far above ω_H.

Magnitude at $\omega = \omega_L/100$

We first evaluate the response at a frequency that is two decades below the lower corner frequency. Substituting $\omega = \omega_L/100$ gives

$$G\left(\frac{\omega_L}{100}\right) = \frac{j\left(\dfrac{\omega_L/100}{\omega_L}\right)}{1 + j\left(\dfrac{\omega_L/100}{\omega_L}\right)} \cdot \frac{1}{1 + j\left(\dfrac{\omega_L/100}{\omega_H}\right)}.$$

The ratio in the first factor is simply

$$\frac{\omega_L/100}{\omega_L} = \frac{1}{100}.$$

For the second factor, use $\omega_H = 100\omega_L$:

$$\frac{\omega_L/100}{\omega_H} = \frac{\omega_L/100}{100\omega_L} = \frac{1}{10{,}000}.$$

Therefore,

$$G\left(\frac{\omega_L}{100}\right) = \frac{j/100}{1 + j/100} \cdot \frac{1}{1 + j/10{,}000}.$$

Taking the magnitude,

$$\left|G\left(\frac{\omega_L}{100}\right)\right| = \frac{1/100}{\sqrt{1 + (1/100)^2}} \cdot \frac{1}{\sqrt{1 + (1/10{,}000)^2}}.$$

Because both denominator correction terms are very close to 1,

$$\left|G\left(\frac{\omega_L}{100}\right)\right| \approx 0.01.$$

Expressed in decibels,

$$20\log_{10}(0.01) = -40 \text{ dB}.$$

Thus, at frequencies far below ω_L, the response is strongly attenuated.

Magnitude at $\omega = 10\omega_L$

Next, consider a frequency one decade above the lower corner frequency:

$$G(10\omega_L) = \frac{j\left(\dfrac{10\omega_L}{\omega_L}\right)}{1 + j\left(\dfrac{10\omega_L}{\omega_L}\right)} \cdot \frac{1}{1 + j\left(\dfrac{10\omega_L}{\omega_H}\right)}.$$

The first ratio becomes

$$\frac{10\omega_L}{\omega_L} = 10,$$

and, since $\omega_H = 100\omega_L$,

$$\frac{10\omega_L}{\omega_H} = \frac{10\omega_L}{100\omega_L} = \frac{1}{10}.$$

Hence,

$$G(10\omega_L) = \frac{j10}{1+j10} \cdot \frac{1}{1+j/10}.$$

Its magnitude is

$$|G(10\omega_L)| = \frac{10}{\sqrt{1+10^2}} \cdot \frac{1}{\sqrt{1+(1/10)^2}}.$$

Numerically,

$$|G(10\omega_L)| = \frac{10}{\sqrt{101}} \cdot \frac{1}{\sqrt{1.01}} \approx 0.99.$$

This value is essentially unity, so in decibels the response is approximately

$$20\log_{10}(0.99) \approx 0 \text{ dB}.$$

Therefore, $\omega = 10\omega_L$ lies in the midband region, where the gain is nearly constant.

Magnitude at $\omega = 100\omega_H$

Finally, evaluate the response at a frequency two decades above the upper corner frequency:

$$G(100\omega_H) = \frac{j\left(\frac{100\omega_H}{\omega_L}\right)}{1+j\left(\frac{100\omega_H}{\omega_L}\right)} \cdot \frac{1}{1+j\left(\frac{100\omega_H}{\omega_H}\right)}.$$

Using $\omega_H/\omega_L = 100$ gives

$$\frac{100\omega_H}{\omega_L} = 100\left(\frac{\omega_H}{\omega_L}\right) = 100(100) = 10{,}000,$$

while the second ratio is

$$\frac{100\omega_H}{\omega_H} = 100.$$

Thus,

$$G(100\omega_H) = \frac{j10{,}000}{1+j10{,}000} \cdot \frac{1}{1+j100}.$$

Taking magnitudes,

$$|G(100\omega_H)| = \frac{10{,}000}{\sqrt{1+10{,}000^2}} \cdot \frac{1}{\sqrt{1+100^2}}.$$

The first factor is approximately 1, whereas the second factor is approximately $1/100$. Therefore,

$$|G(100\omega_H)| \approx 0.01.$$

In decibels,

$$20 \log_{10}(0.01) = -40 \text{ dB}.$$

So, at frequencies far above ω_H, the response is again strongly attenuated.

Summary and interpretation

The three results are

$$\left|G\left(\frac{\omega_L}{100}\right)\right| \approx 0.01 \qquad (-40 \text{ dB}),$$
$$|G(10\omega_L)| \approx 0.99 \qquad (\text{approximately } 0 \text{ dB}),$$
$$|G(100\omega_H)| \approx 0.01 \qquad (-40 \text{ dB}).$$

These values confirm the expected band-pass behavior. The magnitude is small well below the lower corner frequency, nearly unity throughout the midband region, and small again well above the upper corner frequency. This interpretation matches the shape shown in Figure 5.2.

5.2 Treatment of Capacitors Across Frequencies

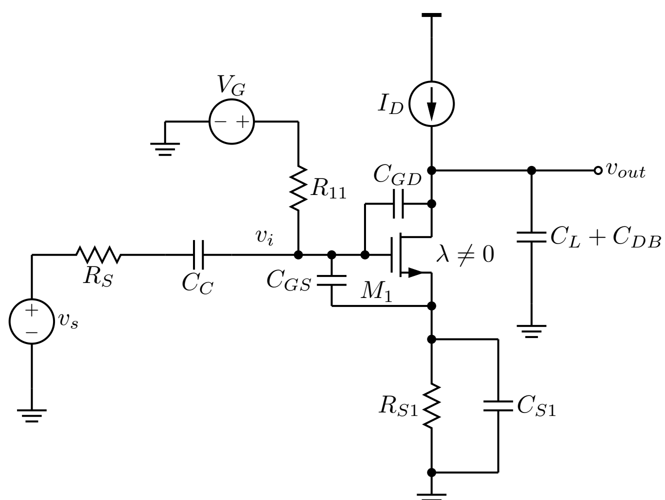

Figure 5.3: A generic common-source amplifier.

When analyzing amplifier circuits over frequency, different capacitors are handled differently depending on whether the circuit is operating in the low-frequency, midband, or high-frequency region. Fig. 5.3 shows a generic amplifier with parastic capacitors and a load capacitor shown explicitly. This section summarizes treatment of capacitors across frequencies.

At low frequencies ($\omega < \omega_L$), the important capacitors are usually the coupling capacitors C_C and bypass capacitors C_B. These capacitors are large, so they cannot be ignored in low-frequency analysis

Type	Example	$\omega < \omega_L$	$\omega_L < \omega < \omega_H$	$\omega > \omega_H$
Coupling Cap	C_C	Kept	Short	Short
Bypass Cap	C_{S1}	Kept	Short	Short
Parasitic Cap	C_{GS},C_{GD}, C_{DB}	Open	Open	Kept
Load Cap	C_L	Open	Open	Kept

Table 5.1: Treatment of various capacitors across frequencies.

and must be kept in the circuit model. In contrast, the much smaller parasitic transistor capacitances C_{GS}, C_{GD}, and C_{DB}, as well as the load capacitance C_L, are treated as open circuits in this region.

In the midband region ($\omega_L < \omega < \omega_H$), the large coupling and bypass capacitors are approximated as short circuits. This is why midband small-signal analysis usually uses a simplified circuit in which these capacitors are replaced by wires. At the same time, the parasitic capacitances and load capacitance are still small enough to be neglected, so they are treated as open circuits.

At high frequencies ($\omega > \omega_H$), the situation reverses. The coupling and bypass capacitors remain effectively short circuits, but the parasitic transistor capacitances C_{GS}, C_{GD}, and C_{DB}, together with the load capacitance C_L, must now be kept in the circuit because they determine the upper-frequency response and the high-frequency cutoff.

The modeling language used in this discussion is straightforward. To keep a capacitor means to leave it in the small-signal schematic. To model it as open means to replace it with an open circuit. To model it as short means to replace it with a short circuit.

In summary, low-frequency analysis keeps large coupling and bypass capacitors, midband analysis shorts them and ignores small parasitic capacitances, and high-frequency analysis keeps the parasitic and load capacitances that shape the upper cutoff. This is summarized in Tab. 5.1.

5.3 *Short-Circuit Time Constant Method*

The short-circuit time-constant method is a standard technique for estimating the low-frequency cutoff, f_L, of an amplifier that contains several coupling and bypass capacitors.

The method begins by disabling all independent sources. Voltage sources are replaced by short circuits, and current sources are replaced by open circuits. This produces the small-signal network used for the time-constant calculation.

Next, one coupling or bypass capacitor is selected at a time. For the capacitor under evaluation, all other coupling and bypass capacitors are replaced by short circuits. At the same time, the load capacitor C_L and the parasitic device capacitances are treated as open circuits. Under these conditions, the equivalent resistance seen by the selected capacitor is found and denoted by R_i.

Once this resistance is known, the corresponding time constant is computed as

$$\tau_i = R_i C_i.$$

Repeating this process for each low-frequency capacitor gives a set of time constants, $\tau_1, \tau_2, \tau_3, \ldots$.

The overall low-frequency cutoff is then estimated from the sum of the reciprocal time constants:

$$f_L \approx \frac{1}{2\pi}\left(\frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{1}{\tau_3} + \cdots\right).$$

This expression shows that the low-frequency response is influenced by all relevant coupling and bypass capacitors, not just by a single capacitor.

A useful interpretation follows from this formula. Since f_L depends on the sum of the terms $1/\tau_i$, the smallest time constant tends to have the greatest effect because it contributes the largest reciprocal term. Thus, the capacitor associated with the smallest τ_i usually has the strongest influence on the low-frequency cutoff.

5.3.1 *Examples*

5.3.1.1 *Common-Source Amplifier*

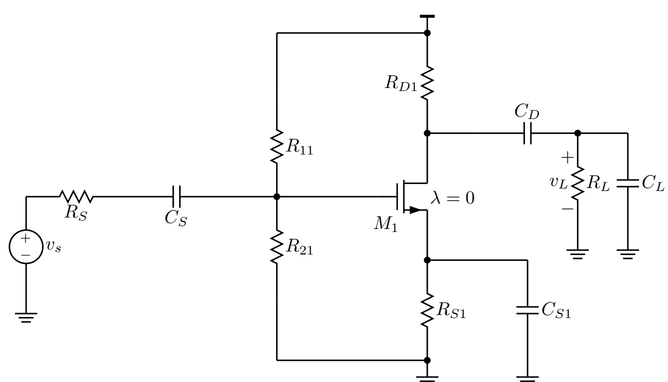

Figure 5.4: A common-source amplifier.

This section presents a step-by-step derivation of f_L for the common-source amplifier in Fig. 5.4 using the short-circuit time-constant method.

Overall idea Before we begin, we set v_s to zero, replace V_{DD} with AC ground, and open the load capacitor, as shown in Fig. 5.5.

For the low-frequency response, the relevant capacitors are the input coupling capacitor C_S, the output coupling capacitor C_D, and

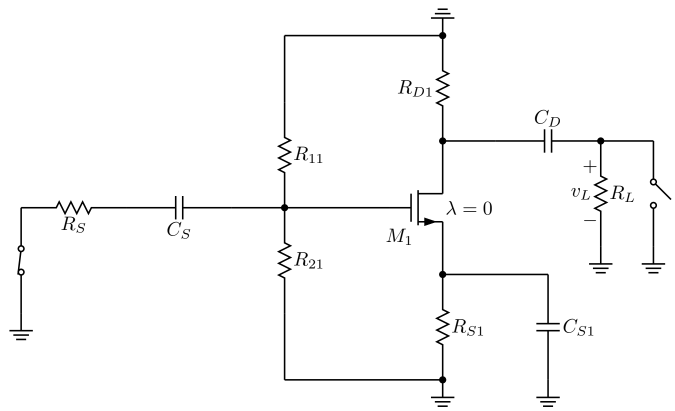

Figure 5.5: Preparation of the common-source amplifier schematic for short-circuit time-constant method analysis.

the source bypass capacitor C_{S1}. Using the short-circuit time-constant method,

$$f_L \approx \frac{1}{2\pi}\left(\frac{1}{\tau_S} + \frac{1}{\tau_D} + \frac{1}{\tau_{S1}}\right).$$

Each time constant is found by selecting one low-frequency capacitor, shorting the other low-frequency capacitors, opening the load capacitor C_L, and then determining the equivalent resistance seen by the selected capacitor.

Determination of τ_S To determine τ_S, keep C_S in the circuit and short the other low-frequency capacitors, C_D and C_{S1}. The load capacitor C_L remains open. The simplified schematic is shown in Fig. 5.6.

With the independent source disabled, the left side of C_S sees the resistance R_S to ground. The right side of C_S is the gate node. Since the MOS gate draws negligible current, that node sees the bias network

$$R_{11} \parallel R_{21}$$

to ground.

Therefore, the resistance seen by C_S is

$$R_{eq,S} = R_S + (R_{11} \parallel R_{21}).$$

Hence,

$$\tau_S = \big(R_S + (R_{11} \parallel R_{21})\big)C_S.$$

Determination of τ_D To determine τ_D, keep C_D in the circuit and short C_S and C_{S1}. Again, C_L remains open. The simplified schematic is shown in Fig. 5.7.

The capacitor C_D lies between the drain side and the load side. Because $\lambda = 0$, we take $r_o = \infty$, so the drain does not provide an additional small-signal path through the transistor. Thus, the left side of C_D sees R_{D1} to ground, while the right side sees R_L to ground.

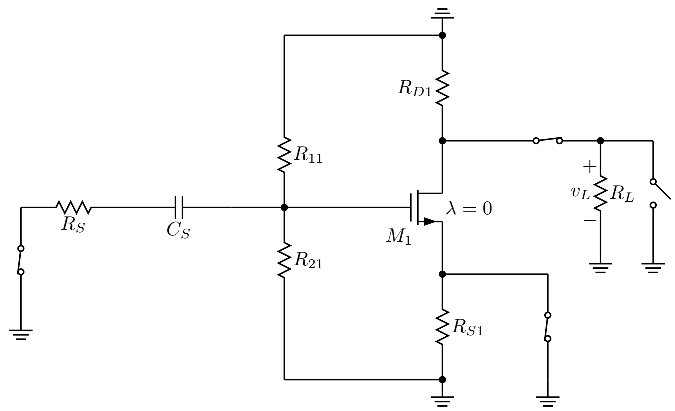

Figure 5.6: Time constant of C_S.

The equivalent resistance seen by C_D is therefore

$$R_{eq,D} = R_{D1} + R_L.$$

So the corresponding time constant is

$$\tau_D = (R_{D1} + R_L)C_D.$$

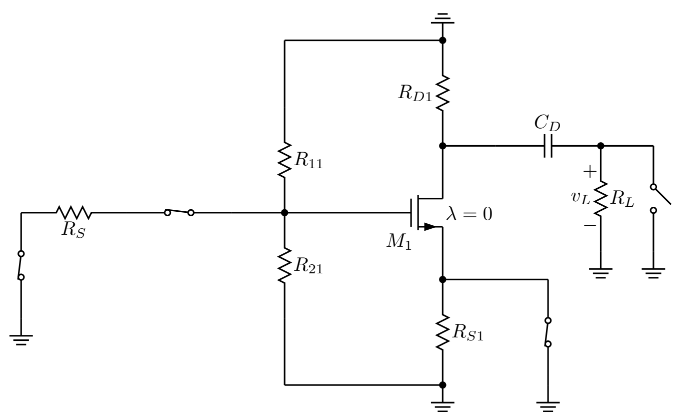

Figure 5.7: Time constant of C_D.

Determination of τ_{S1} To determine τ_{S1}, keep C_{S1} in the circuit and short C_S and C_D. The load capacitor C_L remains open. The simplified schematic is shown in Fig. 5.8.

The capacitor C_{S1} is connected across the source resistor R_{S1}. Looking into the source of the MOS transistor gives the small-signal resistance

$$\frac{1}{g_m}.$$

Therefore, the resistance seen by C_{S1} is

$$R_{eq,S1} = R_{S1} \parallel \frac{1}{g_m}.$$

Hence,

$$\tau_{S1} = \left(R_{S1} \parallel \frac{1}{g_m}\right) C_{S1}.$$

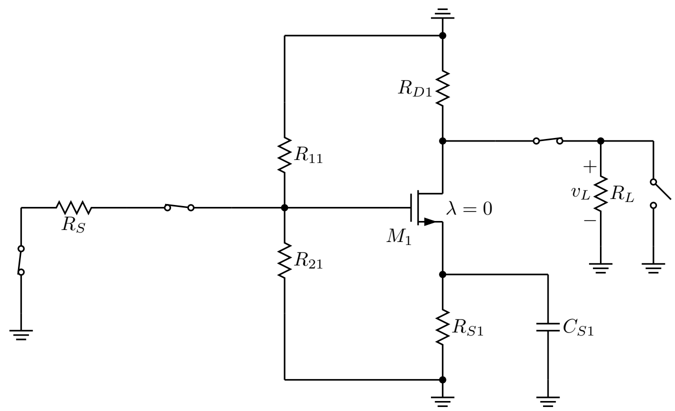

Figure 5.8: Time constant of C_{S1}.

Final result for the low-frequency cutoff Combining the three time constants gives

$$f_L \approx \frac{1}{2\pi}\left[\frac{1}{(R_S + (R_{11} \parallel R_{21}))C_S} + \frac{1}{(R_{D1} + R_L)C_D} + \frac{1}{\left(R_{S1} \parallel \frac{1}{g_m}\right) C_{S1}}\right].$$

Common approximations Two approximations are often used:

- If $R_{11} \parallel R_{21} \gg R_S$, then
$$\tau_S \approx (R_{11} \parallel R_{21})C_S.$$
- If $R_{S1} \gg 1/g_m$, then
$$\tau_{S1} \approx \frac{C_{S1}}{g_m}.$$

Under these approximations,

$$f_L \approx \frac{1}{2\pi}\left[\frac{1}{(R_{11} \parallel R_{21})C_S} + \frac{1}{(R_{D1} + R_L)C_D} + \frac{g_m}{C_{S1}}\right].$$

Furthermore, if $R_{11} \parallel R_{21} \gg R_{D1} + R_L > \frac{1}{g_m}$ and $C_D \approx C_S \approx C_{S1}$, then

$$f_L \approx \frac{1}{2\pi}\frac{g_m}{C_{S1}}.$$

5.3.1.2 *Common-Source Common-Collector Amplifier*

This example explains how to estimate the lower cutoff frequency of a two-stage amplifier made from a common-source MOSFET stage followed by a common-collector BJT stage, shown in Fig. 5.9. The circuit contains four large capacitors that affect the low-frequency response: the input coupling capacitor C_S, the source-bypass capacitor C_{S1}, the interstage coupling capacitor C_C, and the output coupling capacitor C_L. The overall lower cutoff frequency can be estimated by combining the four low-frequency time constants associated with these capacitors.

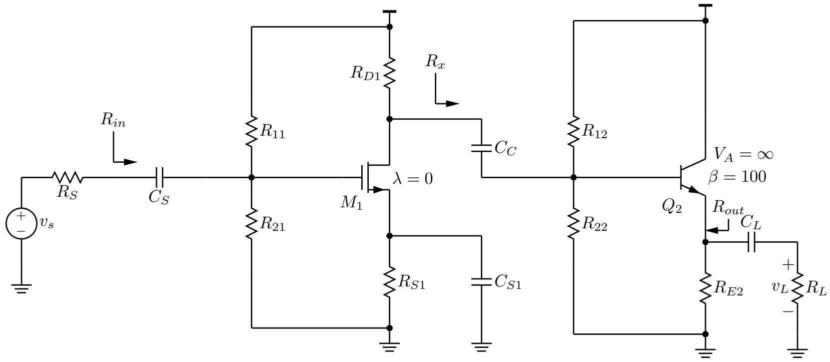

Figure 5.9: A common-source common-collector amplifier.

Overview

The amplifier consists of a common-source first stage and a common-collector second stage. The first stage uses the bias network R_{11} and R_{21}, the drain resistor R_{D1}, and the source resistor R_{S1}. The second stage uses the bias network R_{12} and R_{22} together with the emitter resistor R_{E2} and the load resistor R_L. The coupling capacitors transmit the signal between stages while blocking dc, and the bypass capacitor increases the small-signal gain of the first stage by reducing source degeneration at signal frequencies.

Low-Frequency Approximation

At low frequencies, each large capacitor contributes a high-pass effect. Using the short-circuit time-constant method for the low-frequency case, the lower cutoff frequency is approximated by

$$f_L \approx \frac{1}{2\pi}\left(\frac{1}{\tau_S} + \frac{1}{\tau_{S1}} + \frac{1}{\tau_C} + \frac{1}{\tau_L}\right).$$

Each time constant is found by considering one capacitor at a time while the other large capacitors are treated according to the low-frequency approximation used in the notes.

Individual Time Constants

1. Input Coupling Capacitor C_S

Figure 5.10 shows that C_S sees the source resistance R_S on one side and the input resistance of the first stage on the other side. Since the MOSFET gate current is negligible, the input resistance is approximately the parallel combination of the bias resistors. Therefore,

$$\tau_S = (R_S + R_{11} \parallel R_{21})\, C_S.$$

This term sets the low-frequency effect of the input coupling network.

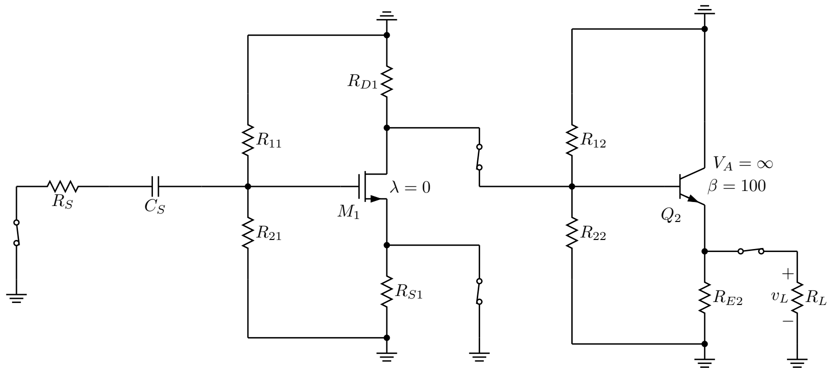

Figure 5.10: Short-circuit time constant associated with C_S.

2. *Source-Bypass Capacitor C_{S1}*

Figure 5.11 shows that C_{S1} is connected across the source resistor of the MOSFET stage. The resistance seen by this capacitor is the parallel combination of R_{S1} and the small-signal resistance looking into the source, which is approximately $1/g_{m1}$. Hence,

$$\tau_{S1} = \left(R_{S1} \parallel \frac{1}{g_{m1}} \right) C_{S1}.$$

This result shows that the bypass capacitor is controlled mainly by the smaller of R_{S1} and $1/g_{m1}$.

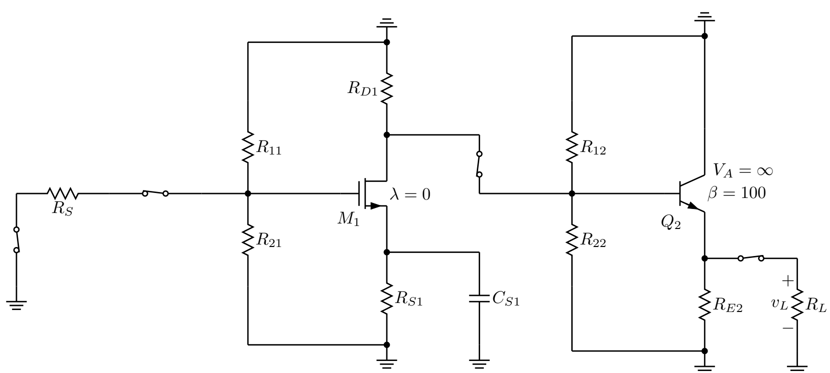

Figure 5.11: Short-circuit time constant associated with C_{S1}.

3. *Interstage Coupling Capacitor C_C*

Figure 5.12 shows that C_C connects the drain of the first stage to the input of the second stage. It therefore sees the output resistance of the first stage and the input resistance of the second stage. The first-stage side contributes approximately R_{D1}, while the second-stage side contributes the parallel combination of the resistance seen looking into the transistor base and the bias network itself. The resulting time constant is

$$\tau_C = (R_{D1} + R_{12} \parallel R_{22} \parallel R_B)\, C_C.$$

Here, R_B denotes the resistance seen looking into the base of the BJT stage; that is,

$$R_B = r_{\pi 2} + (\beta_2 + 1)(R_{E2} \parallel R_L).$$

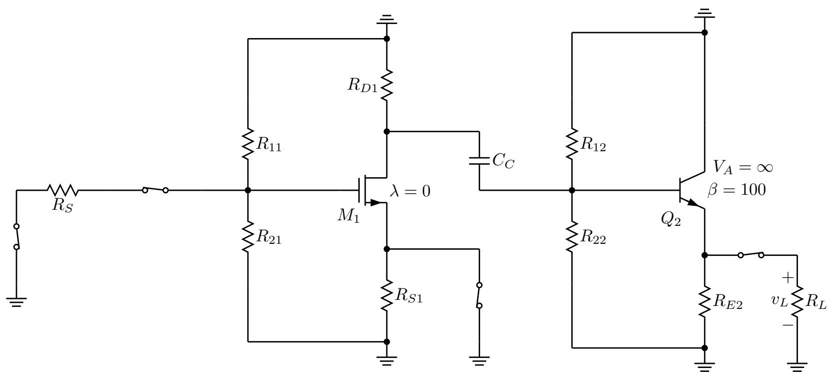

Figure 5.12: Short-circuit time constant associated with C_C.

4. *Output Coupling Capacitor C_L*

Figure 5.13 shows that C_L couples the signal from the emitter follower to the load. On one side, it sees the output resistance of the common-collector stage, which is approximated by $R_{E2} \parallel (1/g_{m2})$. On the other side, it sees the load resistor R_L. The corresponding time constant is

$$\tau_L = \left(R_{E2} \parallel \frac{1}{g_{m2}} + R_L \right) C_L.$$

This term determines how the output coupling capacitor limits low-frequency signal transfer to the load.

Interpretation

The lower cutoff frequency is not determined by only one capacitor. Instead, it depends on the combined effect of all four time constants. A larger time constant produces a smaller corner contribution, and the reciprocal sum of the time constants determines the overall value of f_L. In practice, the capacitor associated with the smallest $1/\tau_i$ contribution has the strongest influence, but all four terms contribute to the final low-frequency bandwidth.

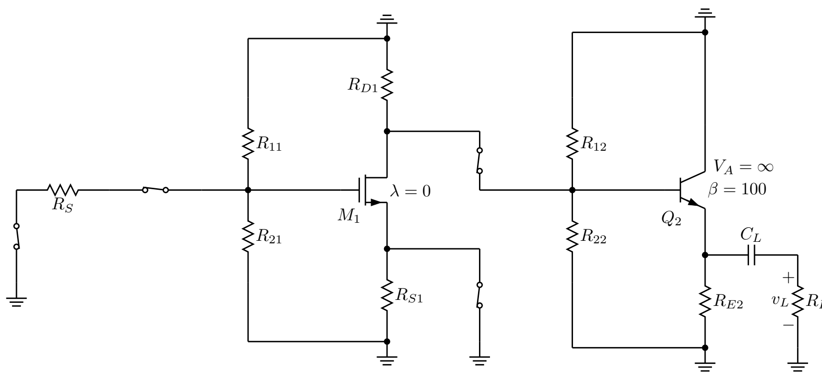

Figure 5.13: Short-circuit time constant associated with C_L.

Summary

This section presents a systematic way to estimate the low-frequency response of a multi-stage amplifier. The procedure is to identify each large capacitor, determine the effective resistance seen by that capacitor, form the corresponding time constant, and then combine the four time constants to obtain the approximate lower cutoff frequency:

$$f_L \approx \frac{1}{2\pi}\left(\frac{1}{\tau_S} + \frac{1}{\tau_{S1}} + \frac{1}{\tau_C} + \frac{1}{\tau_L}\right).$$

This method provides a clear and practical framework for analyzing the low-frequency behavior of cascaded amplifier stages.

5.4 *Open-Circuit Time Constant Method*

The open-circuit time-constant method is a technique used to estimate the upper 3 dB bandwidth, f_H, of an amplifier. The method applies to high-frequency analysis, where the dominant effects come from parasitic and load capacitances rather than from the large coupling and bypass capacitors.

In this approach, the coupling and bypass capacitors are first replaced by short circuits. This step is necessary because the goal is to study the high-frequency behavior of the circuit, and at high frequencies, these large capacitors act approximately as shorts. Any independent voltage sources are then set to zero so that the effective resistance seen by each capacitor can be determined from the resulting small-signal network.

The method proceeds by considering one parasitic or load capacitor at a time. For the capacitor being evaluated, every other parasitic or load capacitor is replaced by an open circuit, while the coupling and bypass capacitors remain shorted. The effective resistance seen by the selected capacitor is denoted by R_i. Once this resistance is found, the corresponding time constant is

$$\tau_i = R_i C_i.$$

This process is repeated for each relevant parasitic or load capacitor in the circuit.

After all the individual time constants have been found, the high-frequency 3 dB bandwidth is approximated from their sum:

$$f_H \approx \frac{1}{2\pi(\tau_1 + \tau_2 + \tau_3 + \cdots)}.$$

This result shows that the upper cutoff frequency is determined by the combined contribution of all relevant capacitances, not by a single capacitor acting alone.

Step-by-Step Procedure

1. Short the coupling and bypass capacitors.

2. Include the relevant parasitic and load capacitors in the small-signal schematic.

3. Set all independent voltage sources to zero.

4. Select one capacitor, C_i, and open all other parasitic and load capacitors.

5. Find the effective resistance, R_i, seen by C_i.

6. Compute the time constant $\tau_i = R_i C_i$.

7. Repeat the process for each remaining parasitic or load capacitor.

8. Sum the time constants and use

   $$f_H \approx \frac{1}{2\pi \sum \tau_i}$$

   to estimate the upper 3 dB frequency.

Interpretation Two key ideas follow from this method. First, the coupling and bypass capacitors are shorted because they are not responsible for the high-frequency roll-off. Second, the quantity that determines f_H is the sum of all the time constants. As a result, the bandwidth is influenced most strongly by the largest contributions to that sum. In practice, a single large time constant may dominate, but the approximation itself is based on adding all of them together.

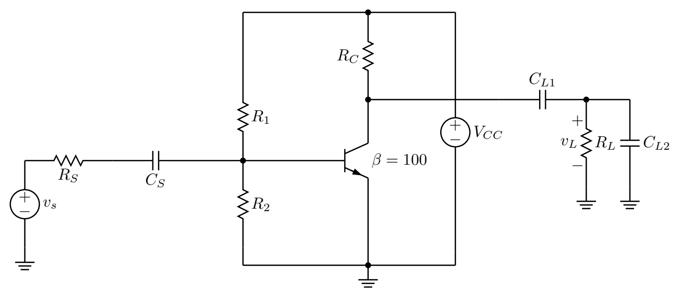

Figure 5.14: A common-emitter amplifier.

5.4.1 Examples

5.4.1.1 Common-Source Amplifier

This example presents a step-by-step derivation of f_H for the common-emitter amplifier shown in Fig. 5.14 using the open-circuit time-constant method. The high-frequency transistor model includes C_π, C_μ, C_{CS}, and the load capacitance.

Overall process The starting point is the open-circuit time-constant approximation

$$f_H \approx \frac{1}{2\pi\left(\tau_\pi + \tau_\mu + \tau_L\right)}.$$

Thus, the task is to determine the three time constants associated with C_π, C_μ, and the output-node capacitance.

To prepare for open-circuit time-constant analysis, v_s is set to zero, V_{CC} is shorted, and coupling capacitors such as C_S and C_{L1} are shorted, as shown in Fig. 5.15. The capacitors C_π, C_μ, and C_{CS} are shown explicitly.

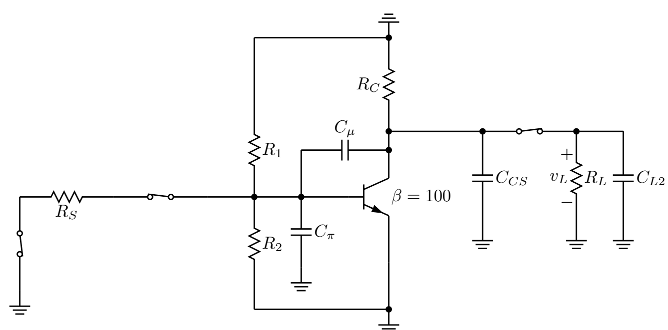

Figure 5.15: Simplified common-emitter amplifier.

Determination of τ_π To find τ_π, keep C_π in the circuit and open the other high-frequency capacitors, as shown in Fig. 5.16. With the independent source set to zero, the base node sees four resistive paths to ground: through R_S, R_1, R_2, and r_π. Therefore, the equivalent resistance seen by C_π is

$$R_\pi = R_S \parallel R_1 \parallel R_2 \parallel r_\pi.$$

The corresponding time constant is

$$\tau_\pi = R_\pi C_\pi = (R_S \parallel R_1 \parallel R_2 \parallel r_\pi)\, C_\pi.$$

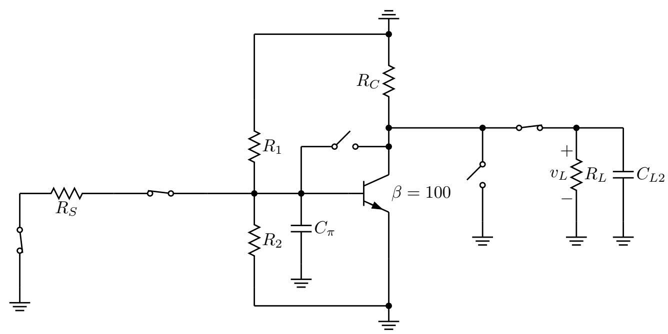

Figure 5.16: Equivalent circuit for calculating τ_π.

Determination of τ_μ Next, consider C_μ, which connects the input node to the collector node. Since this capacitor bridges two active nodes, its equivalent resistance cannot be found by a simple parallel reduction. Let

$$R_B = R_S \parallel R_1 \parallel R_2 \parallel r_\pi, \qquad R_{CEQ} = R_C \parallel R_L.$$

The effective resistance seen by the terminals of C_μ is given by Eq. 3.1.

$$R_\mu = R_B + R_{CEQ} + g_m R_B R_{CEQ}.$$

Hence,

$$\tau_\mu = R_\mu C_\mu = \left(R_B + R_{CEQ} + g_m R_B R_{CEQ}\right) C_\mu.$$

In other words, the resistance seen by C_μ contains an input-side term, an output-side term, and an additional controlled-source contribution.

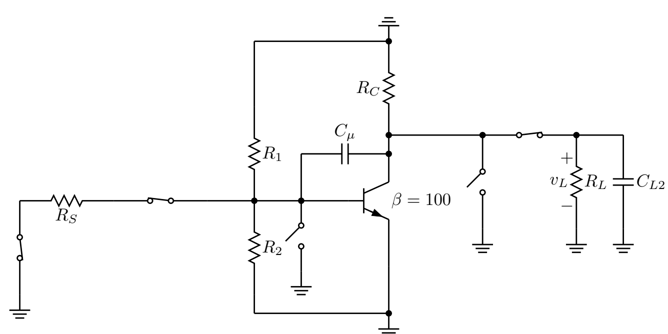

Figure 5.17: Equivalent circuit for calculating τ_μ.

Determination of τ_L The collector, or output, node includes the load-side capacitance together with the transistor output capacitance C_{CS} drawn at the collector node, as shown in Fig. 5.18. Therefore,

$$C_{out} = C_{CS} + C_{L2},$$

and the resistance seen at that node is

$$R_{out} = r_o \parallel R_C \parallel R_L.$$

Therefore,

$$\tau_L = R_{out}C_{out} = (r_o \parallel R_C \parallel R_L)\,(C_{CS} + C_{L2})\,.$$

If the transistor output resistance is assumed to be very large, this reduces to

$$\tau_L \approx (R_C \parallel R_L)\,(C_{CS} + C_{L2})\,.$$

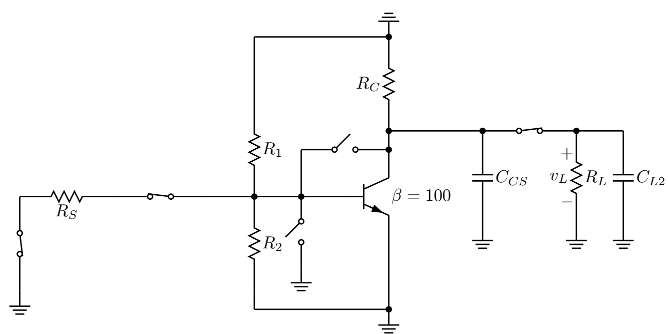

Figure 5.18: Equivalent circuit for calculating τ_L.

Final expression for f_H Substituting the three time constants into the open-circuit time-constant formula gives

$$f_H \approx \frac{1}{2\pi\Big[\,(R_S \parallel R_1 \parallel R_2 \parallel r_\pi)\,C_\pi + \big(R_B + R_{CEQ} + g_m R_B R_{CEQ}\big)\,C_\mu + (r_o \parallel R_C \parallel R_L)\,(C_{CS} + C_{L2})\,\Big]}.$$

Here,

$$R_B = R_S \parallel R_1 \parallel R_2 \parallel r_\pi, \qquad R_{CEQ} = R_C \parallel R_L.$$

With $R_S \ll R_1 \parallel R_2 \parallel r_\pi$, $r_o \gg R_C \parallel R_L$, and $R_S \ll R_{CEQ}$,

$$f_H \approx \frac{1}{2\pi}\frac{1}{R_S C_\pi + (1 + g_m R_S)(R_C \parallel R_L)C_\pi + (R_C \parallel R_L)(C_{CS} + C_{L2})}.$$

5.4.1.2 *Common-Source Common-Collector Amplifier*

This section derives the upper cutoff frequency of a two-stage amplifier made of a common-source MOS stage followed by a common-collector BJT stage shown Fig. 5.19. The analysis uses the open-circuit time-constant method. In this method, all coupling and bypass capacitors are treated as shorts, independent sources are set to zero as shown in Fig. 5.19, and one high-frequency capacitor is considered at a time while all other high-frequency capacitors are opened. The upper 3 dB frequency is then approximated from the sum of the resulting time constants.

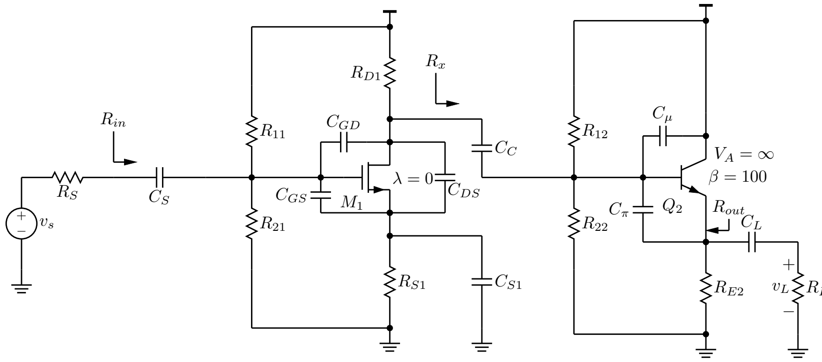

Figure 5.19: High frequency response of a common-source common-collector amplifier.

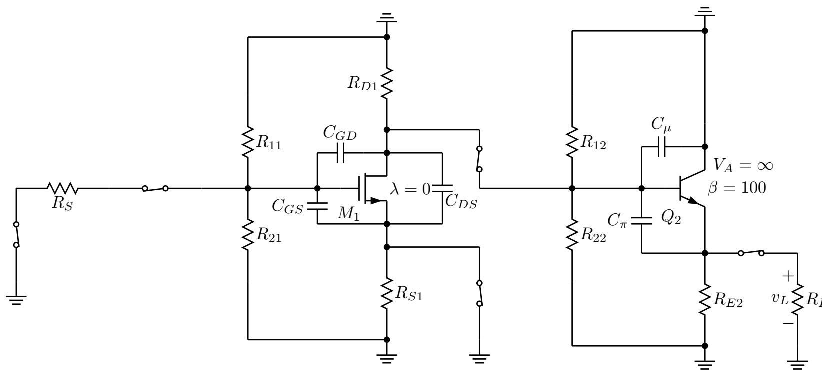

Figure 5.20: Simplified common-source common-collector amplifier for high frequency response analysis.

For the amplifier studied here, the relevant high-frequency capacitors are the MOS capacitances C_{GS}, C_{GD}, and C_{DS}, together with the BJT capacitances C_μ and C_π. The overall result is

$$f_H \approx \frac{1}{2\pi\left(\tau_{GS} + \tau_{GD} + \tau_{DS} + \tau_\mu + \tau_\pi\right)}.$$

Thus, the high-frequency bandwidth is determined by the combined effect of all important device capacitances rather than by a single capacitor alone.

Time Constants Associated with the MOS Stage

Figure 5.21 show that the gate-to-source capacitance C_{GS} sees the resistance connected to the gate node when the source is grounded for small-signal high-frequency analysis. Its time constant is therefore

$$\tau_{GS} = (R_S \parallel R_{11} \parallel R_{21})\, C_{GS}.$$

The gate-to-drain capacitance C_{GD} is more important because it connects the input and output sides of the MOS transistor. Figure

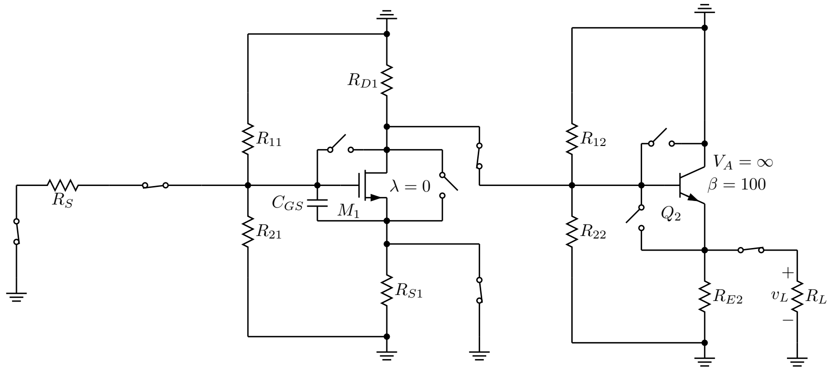

Figure 5.21: Equivalent circuit for finding τ_{GS}.

5.22 shows the equivalent circuit for evaluating τ_{GD}.

$$\tau_{GD} = (R_{D1} \parallel R_x + (1 + g_{m1}(R_{D1} \parallel R_x))\, R_S)\, C_{GD},$$

where

$$R_x = R_{12} \parallel R_{22} \parallel R_B, \qquad R_B = r_\pi + (\beta + 1)(R_{E2} \parallel R_L).$$

Since the second-stage input resistance is typically much larger than R_{D1}, the approximation

$$\tau_{GD} \approx (R_{D1} + (1 + g_{m1} R_{D1}) R_S)\, C_{GD}$$

is used. This shows that C_{GD} is strongly affected by gain-related resistance multiplication.

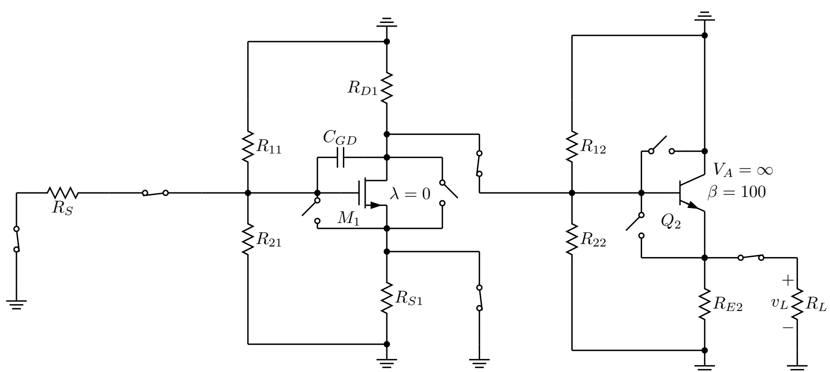

Figure 5.22: Equivalent circuit for finding τ_{GD}.

Figure 5.23 shows the equivalent circuit which gives rise to τ_{DS}. The drain-to-source capacitance C_{DS} contributes a simpler time constant:

$$\tau_{DS} = (R_{D1} \parallel R_x) C_{DS} \approx R_{D1} C_{DS}.$$

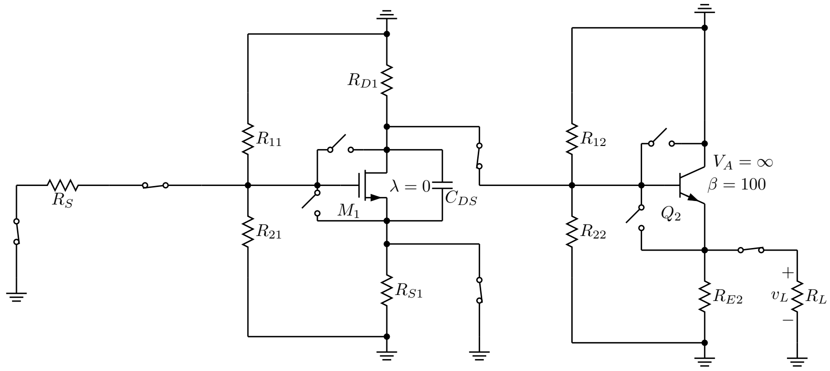

Figure 5.23: Equivalent circuit for finding τ_{DS}.

Effective Resistance Used in the BJT Analysis

Before finding the BJT time constants, recall that the effective resistance seen between the base and emitter side of the transistor is given by Eq. 2.52.

$$R_{\text{eff}} = \left(\frac{R_B + R_E}{1 + g_m R_E} \right) \parallel r_\pi.$$

Time Constants Associated with the BJT Stage

Figure 5.24 shows the equivalent circuit which gives rise to the base-collector time constant.

$$\tau_\mu = (R_{D1} \parallel R_x) C_\mu \approx R_{D1} C_\mu,$$

with the same resistance definition

$$R_x = R_{12} \parallel R_{22} \parallel R_B, \qquad R_B = r_\pi + (\beta + 1)(R_{E2} \parallel R_L).$$

Because the resistance looking into the second stage is large, the approximation to $R_{D1} C_\mu$ is appropriate.

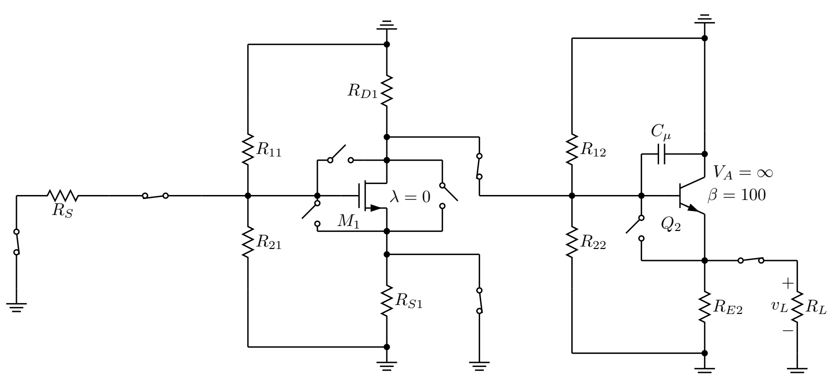

Figure 5.24: Equivalent circuit for finding τ_μ.

The base-emitter capacitance C_π uses the effective input resistance of the BJT stage. Figure 5.25 shows the equivalent circuit which gives rise to τ_π.

$$\tau_\pi \approx \left(r_\pi \parallel \frac{R_{D1} + R_L}{1 + g_{m2} R_L} \right) C_\pi.$$

Equivalently, this can be interpreted in terms of the effective resistance obtained from the base-emitter test-source calculation.

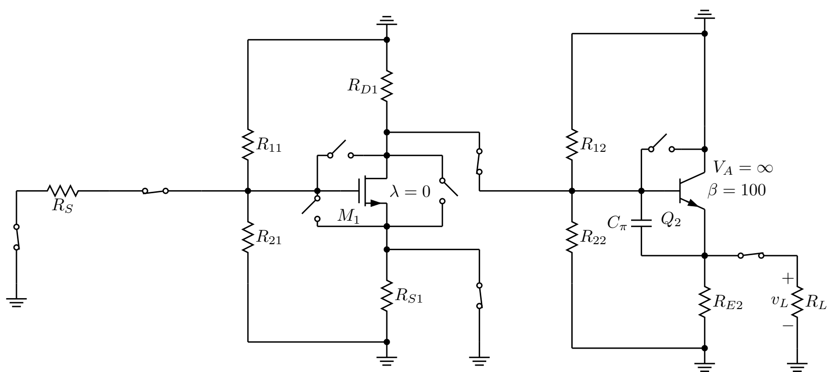

Figure 5.25: Equivalent circuit for finding τ_π.

Interpretation of the Result

The main idea of the section is that the upper cutoff frequency of a multi-stage amplifier can be estimated by summing the open-circuit time constants of all relevant high-frequency capacitors. Capacitances internal to the MOS transistor contribute through τ_{GS}, τ_{GD}, and τ_{DS}, while the BJT contributes through τ_μ and τ_π. Among these terms, the capacitors that bridge input and output nodes, especially C_{GD}, often make the largest contribution because the resistance seen by such capacitors can be magnified by transistor gain.

In summary, the section shows how a complicated multi-stage high-frequency problem can be reduced to a sequence of smaller resistance calculations, each tied to a single capacitor. Once these time constants are found, their sum provides a practical estimate of the amplifier's high-frequency 3 dB bandwidth.